

Assessment of the South African anchovy resource using data up to 2024

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The baseline assessment of the South African anchovy resource has been updated to include data from 2024. This model, which assumes time-invariant natural mortality, is compared with time-varying and density dependant natural mortality alternatives. Given the recent consecutive years of poor recruitment, one alternative to a time-invariant Beverton Holt stock recruitment relationship is also re-considered.

Keywords: anchovy, assessment, natural mortality, population dynamics, South Africa

Introduction

de Moor (2024) updated the assessment of the South African anchovy resource, using data up to 2023, and reconsidered model assumptions to develop a new 'benchmark' assessment. This document updates the data to 2024 and considers a few further tests highlighted both in de Moor (2024) and in recent discussions of the Small Pelagic Scientific Working Group and its Task Groups.

Methods

South African anchovy are assumed to form a single stock distributed along the west and south coasts of South Africa. The integrated assessment is age structured, fitting to length structured data. The population dynamics model is detailed in Appendix B and the survey and commercial data available for this assessment from 1984 to 2024 have been detailed in de Moor *et al.* (2024b) and Appendix A. The model parameters are defined in Tables B.1 and B.2. Key methodological changes since the previous benchmark assessment by de Moor (2020a,b) include:

- i) Model quarters are now January-March, April-June, July-September and October-December, instead of November – January, February – April, May – July, August - October.
- ii) Spawning and recruitment are assumed to occur at the start of the 4th quarter, 1st October, instead of 1st November. October has been a peak month for anchovy spawning out of an October-January spawning season as observed in 1977-78 the Cape Egg and Larval Program (CELP) and 1993/4-1994/5 SARP surveys (Painting and Korrûbel 1998; van der Lingen and Huggett 2003), although December was found to be the peak month in ichthyoplankton surveys in the 1960s and Sardine and Anchovy Recruitment Programme (SARP) line from 1995-2001 (Mullon *et al.* 2003).
- iii) The November hydro-acoustic survey is assumed to provide an estimate of relative abundance corresponding to the total biomass at 15th November (Figure A.1 of de Moor 2022), rather than 1st November.

In updating this assessment, a newly available prior distribution for the November survey bias has been incorporated, and the model has been adjusted to account for the delayed November hydroacoustic survey of total biomass in the most recent two years. The results of de Moor (2024) gave support to the baseline model assuming a time-invariant growth curve and stock-recruitment relationship, based off results assuming time-invariant natural mortality. This document updates this possible baseline model and re-considers time-varying natural mortality together with density dependent natural mortality following

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requests by the Ecosystem Inputs to Management Task Group of the Small Pelagic Scientific Working Group. In addition, given the recent consecutively low recruitment years the option with two Beverton Holt stock recruitment relationships has also been updated:

- A_{BH} - Beverton Holt stock-recruitment relationship, time invariant natural mortality
- A_{3BH} - two Beverton Holt stock-recruitment curves, one estimated from 1984 to 1998 and 2013 to 2024, and the other from 1999 to 2012.
- A_{Mad} – annually varying adult natural mortality, i.e. random effects model with $\sigma_{ad} = 0.2$ or $\sigma_{ad} = 0.1$, and $\rho \sim U(0,1)$.
- A_{Mj} – annually varying juvenile natural mortality, i.e. random effects model with $\sigma_j = 0.2$ or $\sigma_j = 0.1$, and $\rho \sim U(0,1)$.
- A_{DDM1} - density dependent natural mortality, increasing at low biomass to account for the greater relative predation on forage fish when their biomass is low (e.g. Saraux *et al.* (2021)): $\bar{M}_{j,y} = \bar{M}_{ad,y} = \bar{M} + e^{-\chi B_{y-1}}$.
- A_{DDM2} - density dependent natural mortality, additionally increasing at high biomass to account for the possibility of competition for food when biomass reaches carrying capacity: $\bar{M}_{j,y} = \bar{M}_{ad,y} = \bar{M} + e^{-\chi B_{y-1}} + \max\{0, e^{-\chi_2(B_{y-1} - 4500)} - 1\}$.
- A_{DDM3} - density dependent natural mortality, additionally increasing at high biomass to account for the possibility of competition for food when biomass reaches carrying capacity: $\bar{M}_{j,y} = \bar{M}_{ad,y} = \bar{M} + e^{-\chi B_{y-1}} + \max\{0, \chi_2(B_{y-1} - 4500)\}$.

The model estimated November biomass depends on the subsequent year's natural mortality for the 4th quarter (Equation B.5). It should therefore be noted that in the density dependent natural mortality scenarios, the constant $\bar{M}_j = \bar{M}_{ad} = 1.2$ is assumed for this calculation only.

Results and Discussion

Results are given at the joint posterior mode only. Figures 1 to 12 show the model fit to the data and associated key estimated relationships for A_{BH}. The model fit to the hydroacoustic survey data and DEPM estimates is relatively good (Figures 1 to 3), with the model estimating lower biomass in 2020-2023 than that estimated by de Moor (2024) (Figures 1 to 3).

Figures 5 and 7 indicate that the model fits to the proportion-at-length data are good on average. However, the fits are poorer in the most recent years (residuals in Figures 5,7). The differences in the survey length frequencies in the most recent 2 years (Appendix C) are likely due to a reduction in adult anchovy relative to juvenile anchovy following consecutive years of very poor recruitment. However, the impact of the different time for the survey is not fully known. While catches in quarter 1 are typically small, given the low biomass in recent years, equation (B.5) might need to be updated to take quarter 1 catches into account.

While the model structure estimates recruitment at 1st October for consistency reasons, as both sardine and anchovy models will be combined in the Operating Model for the next Management Strategy Evaluation, the effective timing of peak recruitment as estimated by the model is approximately early-December (Figure 8, $t_0 = 0.19$), consistent with de Moor (2024). This corresponds with approximately the middle of the October-January spawning season (Painting and Korrûbel 1998; Mullon *et al.* 2003; van der Lingen and Huggett 2003).

The estimated Beverton-Holt stock recruitment relationship is shown in Figure 10 and the timeseries of November recruitments in Figure 11 (noting recruitment in the final year may not be well estimated given the data available; recruitment in October 2023 for A_{BH} is substantially less than that estimated by de Moor (2024)). While there appears to be no trend in residuals against spawner biomass, there is a decreasing trend in recruitment residuals in recent years.

When two Beverton-Holt stock recruitment relationships are assumed (A_{3BH}), the fit to the November survey data is slightly worse, particularly in 2000-2002, 2011, 2024 (Table 1, Figure 1), but this is offset by an improved fit to the other data and a substantial reduction in the recruitment residuals (Table 1, Figure 10b). The curve estimated for the early and later years is substantially less productive than that estimated by A_{BH} . This alternative relationship could have a substantial impact on projections with the Management Strategy Evaluation.

Figure 13 shows the annual estimated juvenile and adult natural mortality rates when these are (separately) allowed to vary interannually (A_{Mj} and A_{Mad}) while assuming a time-invariant Beverton Holt stock recruitment relationship. When $\sigma_j = 0.1$, the natural mortality is estimated to be highly correlated, with $\rho = 0.86$ and $\rho = 0.91$ for A_{Mj} and A_{Mad} , respectively. This is a substantial increase in correlation – and trend – since that estimated by de Moor (2024). When $\sigma_j = 0.2$, the adult natural mortality rate is estimated similarly to de Moor (2024) with $\rho = 0.59$. There is little improvement in the fit to the data for A_{Mj} , but a substantial improvement is obtained with A_{Mad} (Table 1).

Allowing for density dependent natural mortality results in a small improvement in the fit to the data (Table 1). In these models, the increase in natural mortality as biomass tends to zero is constrained to 1. An alternative that was tested, estimating this maximum increase, resulted in a very (unrealistic) large increase being estimated as biomass tends to zero. The equation forms currently used for A_{DDM2} and A_{DDM3} allow for unconstrained increases in natural mortality for high biomass. If these are to be further pursued, alternative equations may need to be investigated, in addition to the current (arbitrary) threshold of 4.5 million t, above which natural mortality is estimated to increase.

Summary and Next Steps

While A_{BH} should be a straightforward update to the baseline model of de Moor (2024), given the availability of another year's data, these results show that the consecutively low recruitments in recent years may require some refinement to stock-recruitment assumptions for the baseline Operating Model(s) for the joint sardine-anchovy Management Strategy Evaluation.

The draft set of robustness tests has been updated as follows:

Higher priority

- A_M - alternative fixed natural mortality (to be finalised). Improved fits to the data are obtained for higher values (de Moor 2024), particularly for increases in $\bar{M}_j$. However, a lower value might still be a plausible alternative. Suggestions: (i) $\bar{M}_j = 1.5$ and $\bar{M}_{ad} = 1.2$ and (ii) $\bar{M}_j = \bar{M}_{ad} = 0.9$.
- A_{HS} - hockey stick stock-recruitment curve, with uniform priors on the log of the maximum recruitment and on the ratio of the spawning biomass at the inflection point to carrying capacity
- A_{3HS} - two hockey stick stock-recruitment curves, with uniform priors on the log of the maximum recruitment and on the ratio of the spawning biomass at the inflection point to carrying capacity, one estimated using data from 1984 to 1998 and 2013 to 2024, and the other from 1999 to 2012.
- A_{3BH} - two Beverton Holt stock-recruitment curves, with uniform priors on steepness and carrying capacity, one estimated using data from 1984 to 1998 and 2013 to 2024, and the other from 1999 to 2012

In cases where a second curve is estimated from 2000 to 2012 or 2024, the variance about the stock recruitment curve over this time period, $\sigma_{r,2000+}^2$, is estimated separately from that for the earlier time period, σ_r^2 .

Lower priority

- A_R** - Ricker stock-recruitment curve, with uniform priors on steepness and carrying capacity
- A_{DDM}** - potential modifications to the density dependent natural mortality options considered in this document.
- A_{t0}** - annually varying t_0 , allowing for inter-annual variability in the timing of peak spawning. Equations B.6 and B.7 are adjusted as follows:

$$A_{y,a,l}^{Oct} \sim N \left(L_{\infty} \left(1 + (D-1)e^{-\kappa(a-t_{0,y}-a)} \right)^{1/(1-D)}, \vartheta_a^2 \right) \quad 0 \leq a \leq 4^+, 1.5^- \text{ cm} \leq l \leq 16^+ \text{ cm}$$

$$A_{y,q,a,l} \sim \begin{cases} N \left(L_{\infty} \left(1 + (D-1)e^{-\kappa(a+(2q+1)/8-t_{0,y}-a)} \right)^{1/(1-D)}, \left(1 - \frac{2q+1}{8} \right) \vartheta_a^2 + \frac{2q+1}{8} \vartheta_{a+1}^2 \right) & 1 \leq q \leq 3 \\ N \left(L_{\infty} \left(1 + (D-1)e^{-\kappa(a+1/8-t_{0,y-1}-a)} \right)^{1/(1-D)}, \left(1 - \frac{1}{8} \right) \vartheta_a^2 + \frac{1}{8} \vartheta_{a+1}^2 \right) & q = 4 \end{cases}$$

$$0 \leq a \leq 4^+, 1.5^- \text{ cm} \leq l \leq 16^+ \text{ cm}$$

with $t_{0,y} = t_0 + \varepsilon_y^t$

$$\text{and } \varepsilon_y^t = \begin{cases} \eta_y^t & y = y_0 \\ \rho^t \varepsilon_{y-1}^t + \sqrt{1 - (\rho^t)^2} \eta_y^t & y_1 < y \leq y_n \end{cases}$$

t_0 - average age at which the length (in expectation) is zero ($t_0 = \frac{1}{\kappa_c} \left\{ \ln \left(\frac{L_{c,1} - L_{c,\infty}}{L_{c,\infty}} \right) + \kappa_c \right\}$, where $L_{c,a=1} \sim U(5,18)$)

ε_y^t - annual autocorrelated residuals about the age at which length is zero

$\eta_y^t \sim N(0, 0.2^2)$ - annual uncorrelated residuals about the age at which length is zero

$\rho^t \sim U(-1,1)$ - autocorrelation coefficient in these residuals

- A₄** - No plus group, all remaining fish assumed to die as they reach age 5.
- A_{sur}** - Survey selectivity below 7.5¹cm was estimated to be a constant, and uniform (1) selectivity was assumed for lengths ≥ 7.5 cm.
- A_{com}** - Commercial selectivity was not estimated to decrease at higher lengths, i.e. $\delta_q = 0$.
- A_{com2}** - Commercial selectivity was modelled using a double-logistic curve.
- A_{sel}** - commercial selectivity-at-length is estimated separately for 1984-2003 and 2004-2023, to account for a greater proportion of anchovy catch been taken south/east of Cape Point since ~2003 (Coetzee and Merkle 2024) where larger/older fish are found.
- A_{kegg}** - Egg survey bias estimated with uninformative prior, i.e. $\ln(k_g^A) \sim U(-100, 0.7)$.
- A_{kegg1}** - Negatively biased egg surveys, i.e., $k_g^A = 0.75$.
- A_{kegg2}** - Positively biased egg surveys, i.e., $k_g^A = 1.25$.
- A_{lamR}** - Additional variance (over and above the survey sampling CV) associated with the recruit survey fixed $(\lambda_r^A)^2 = 0$.
- A_{lamN}** - Additional variance (over and above the survey sampling CV) associated with the November survey, estimated with $(\lambda_N^A)^2 \sim U(0, 100)$.
- A_{lamN2}** - Additional variance (over and above the survey sampling CV) associated with the November survey fixed $(\lambda_N^A)^2 = 0.02$.
- A_{growth}** - Richards growth curve rather than von Bertalanffy growth curve.
- A_{like}** - Multinomial likelihood for proportion at length data.
- A_{mat}** - Time-invariant maturity ogive.

¹ This length was chosen as it provided the best deterministic fit in the range 6-9cm.

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Table 1. The contributions to the negative log posterior pdf at the joint posterior mode for the alternative models considered in this document.

			-lnLikelihood					-lnPriors					Survey bias		
	$-\ln Posterior$	$-\ln L$	$-\ln L^{Nov}$	$-\ln L^{Egg}$	$-\ln L^{rec}$	$-\ln L^{surp}$	$-\ln L^{com}$	ε_y	All growth parame- ters	δ_q	$N_{1983,a}$	η_y^j	k_N	k_N	k_r
A _{BH}	-795.62	-859.24	1.62	5.12	36.70	-546.01	-356.68	45.19	4.50	0.50	13.36	0.00	-0.06	0.93	0.66
A _{3BH}	-801.00	-857.39	4.63	5.09	36.15	-546.48	-356.78	37.89	4.60	0.50	13.36	0.00	-0.09	0.92	0.65
A _{Mj} with $\sigma_j = 0.2$	-827.52	-858.41	1.45	5.28	36.47	-545.46	-356.14	37.70	4.50	0.50	13.36	-25.25	-0.07	0.93	0.66
A _{Mj} with $\sigma_j = 0.1$	-854.17	-859.04	1.50	5.14	36.46	-546.05	-356.10	43.80	4.47	0.50	13.36	-57.33	-0.07	0.93	0.67
A _{Mad} with $\sigma_{ad} = 0.2$	-831.69	-872.93	1.27	5.80	31.92	-556.78	-355.14	45.87	4.67	0.50	13.43	-23.28	-0.12	0.90	0.63
A _{Mad} with $\sigma_{ad} = 0.1$	-856.13	-863.78	1.07	5.55	33.93	-548.26	-356.05	45.23	4.53	0.50	13.40	-56.08	-0.09	0.92	0.64
A _{DDM1}	-795.68	-859.32	1.35	5.11	36.92	-545.79	-356.91	45.20	4.51	0.50	13.36	0.00	-0.06	0.94	0.66
A _{DDM2}	-797.20	-862.79	0.84	5.17	36.44	-548.78	-356.46	47.10	4.55	0.50	13.36	0.00	-0.06	0.93	0.65
A _{DDM3}	-796.67	-860.77	1.12	5.19	36.51	-546.81	-356.78	45.71	4.53	0.50	13.36	0.00	-0.13	0.89	0.63

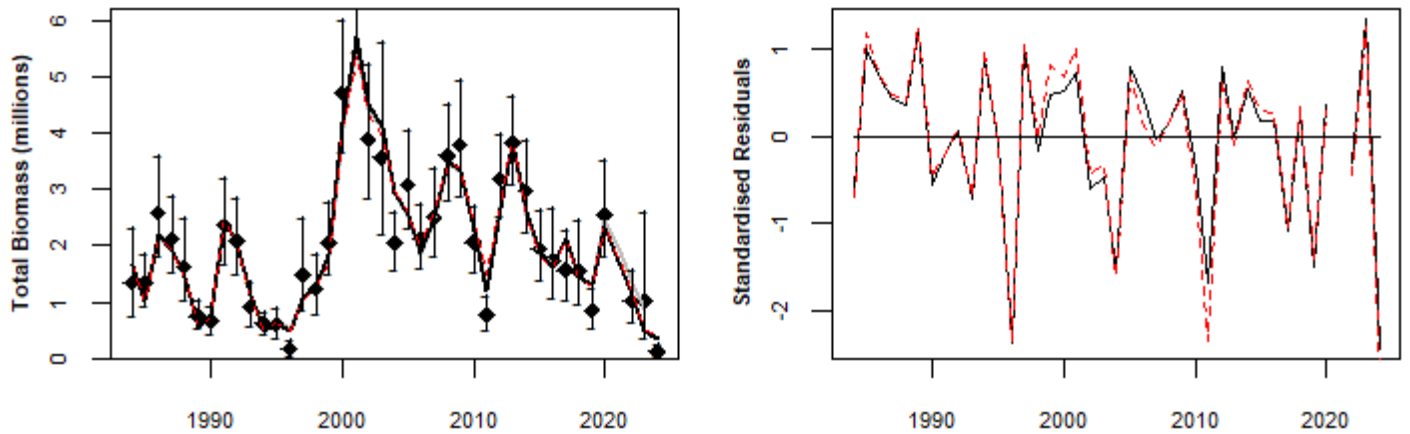


Figure 1. Acoustic November survey and bias corrected model estimated anchovy biomass from 1984 to 2024 for A_{BH} (black) and A_{3BH} (red); the 2023 and 2024 values correspond to that predicted and surveyed in Feb-Mar 2024 and Mar-Apr 2025, respectively. The survey indices are shown with 95% confidence intervals reflecting survey inter-transect variance. The standardised residuals are given in the right-hand plot. The A_{BH} November anchovy biomass estimated by de Moor (2024) is shown in grey.

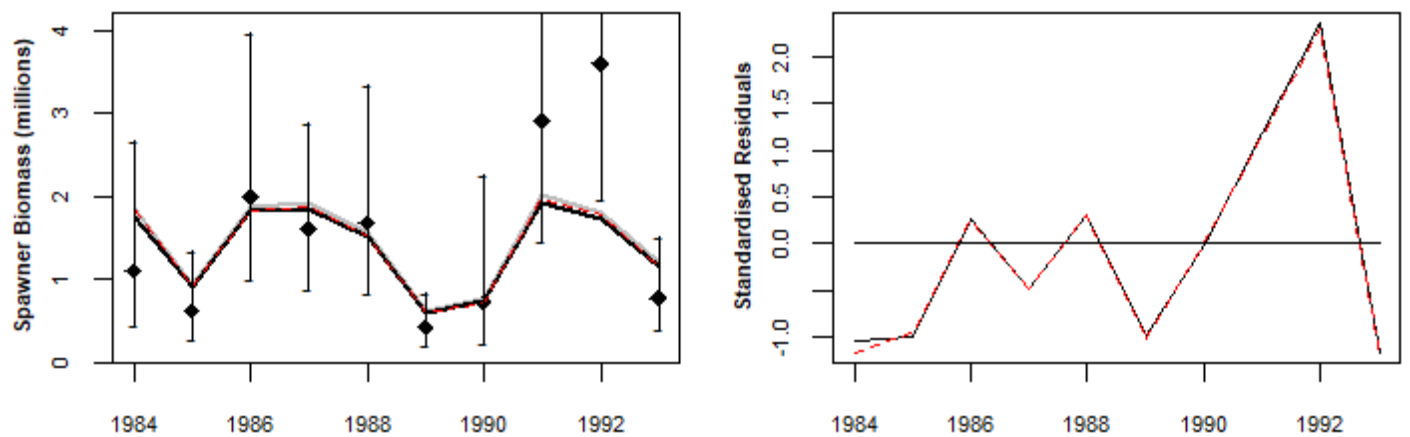


Figure 2. Daily egg production method and bias corrected model estimated anchovy spawner biomass from 1984 to 1993 for A_{BH} (black) and A_{3BH} (red). The survey indices are shown with 95% confidence intervals. The standardised residuals are given in the right-hand plot. The A_{BH} anchovy spawner biomass estimated by de Moor (2024) is shown in grey.

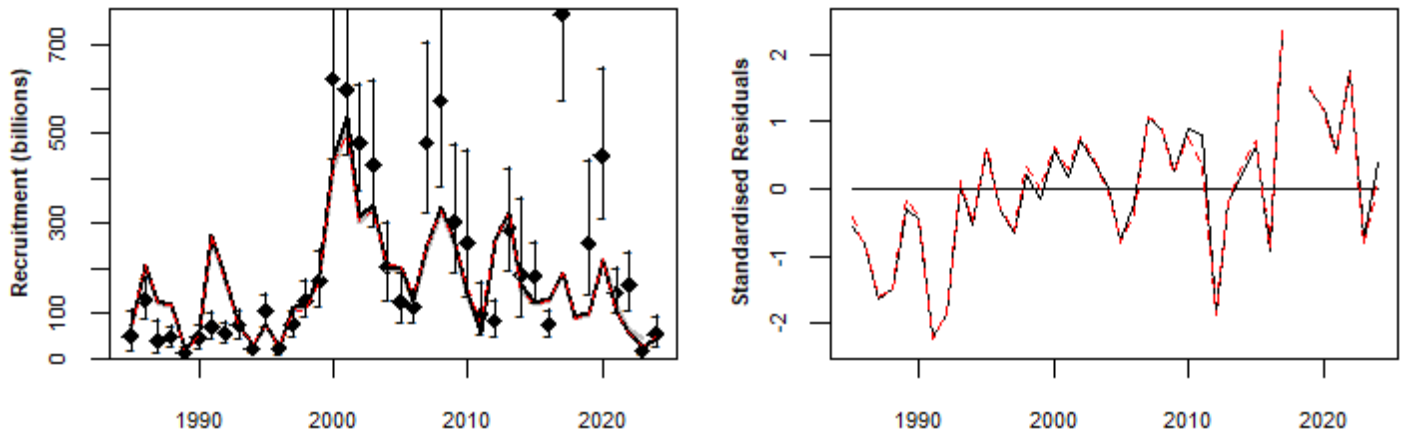


Figure 3. Acoustic survey and bias corrected model estimated anchovy recruitment numbers from 1985 to 2024 for A_{BH} (black) and A_{3BH} (red). The survey indices are shown with 95% confidence intervals reflecting survey inter-transect. Note that additional survey variance is estimated included in the model, and not shown in this figure. The standardised residuals are given in the right-hand plot. The A_{BH} anchovy recruitment estimated by de Moor (2024) is shown in grey.

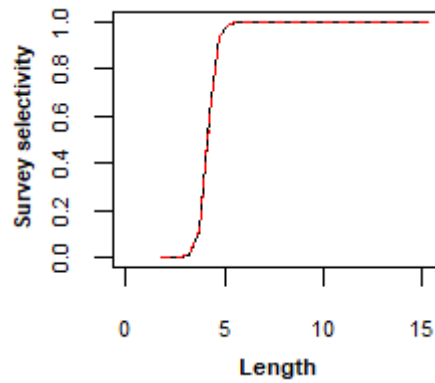


Figure 4. Model estimated trawl survey selectivity at length for A_{BH} (black) and A_{3BH} (red).

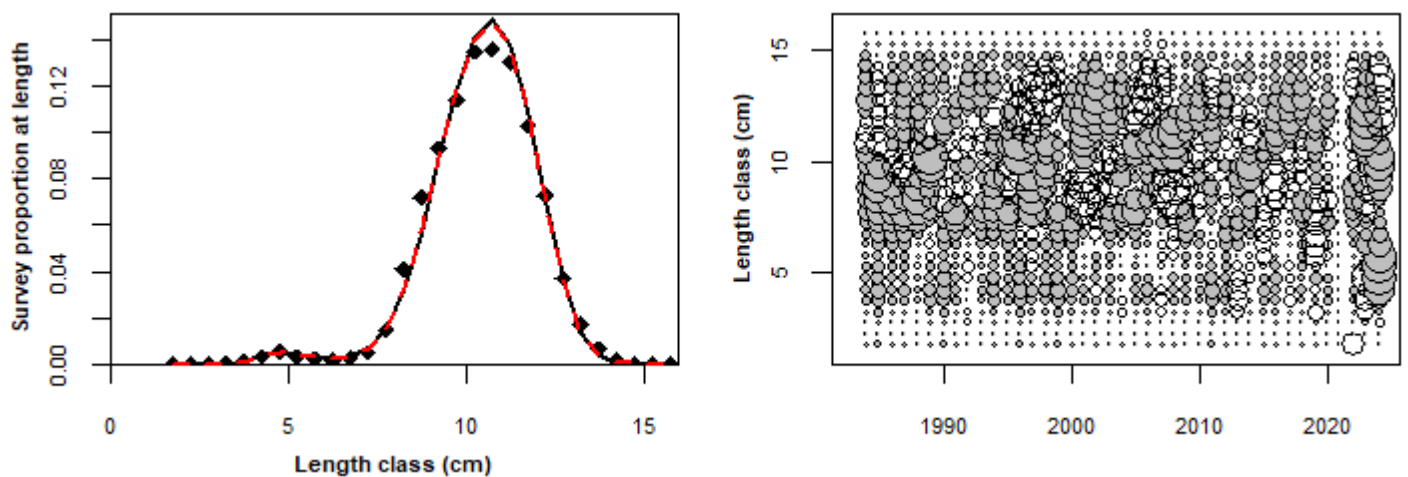


Figure 5. Average (over all years) model predicted and observed proportions-at-length in the November survey trawls for A_{BH} (black) and A_{3BH} (red). The standardised residuals for A_{BH} are given in the bubble plot; the size of the bubbles are proportional to the absolute value of the residuals, while the shaded bubbles show negative and the unshaded bubbles show positive residuals.

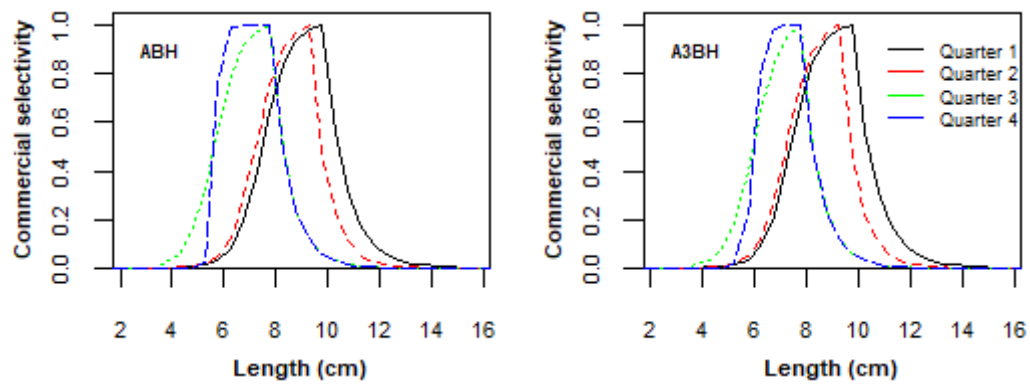


Figure 6. Model estimated quarterly commercial selectivity at length for A_{BH} (left) and A_{3BH} (right).

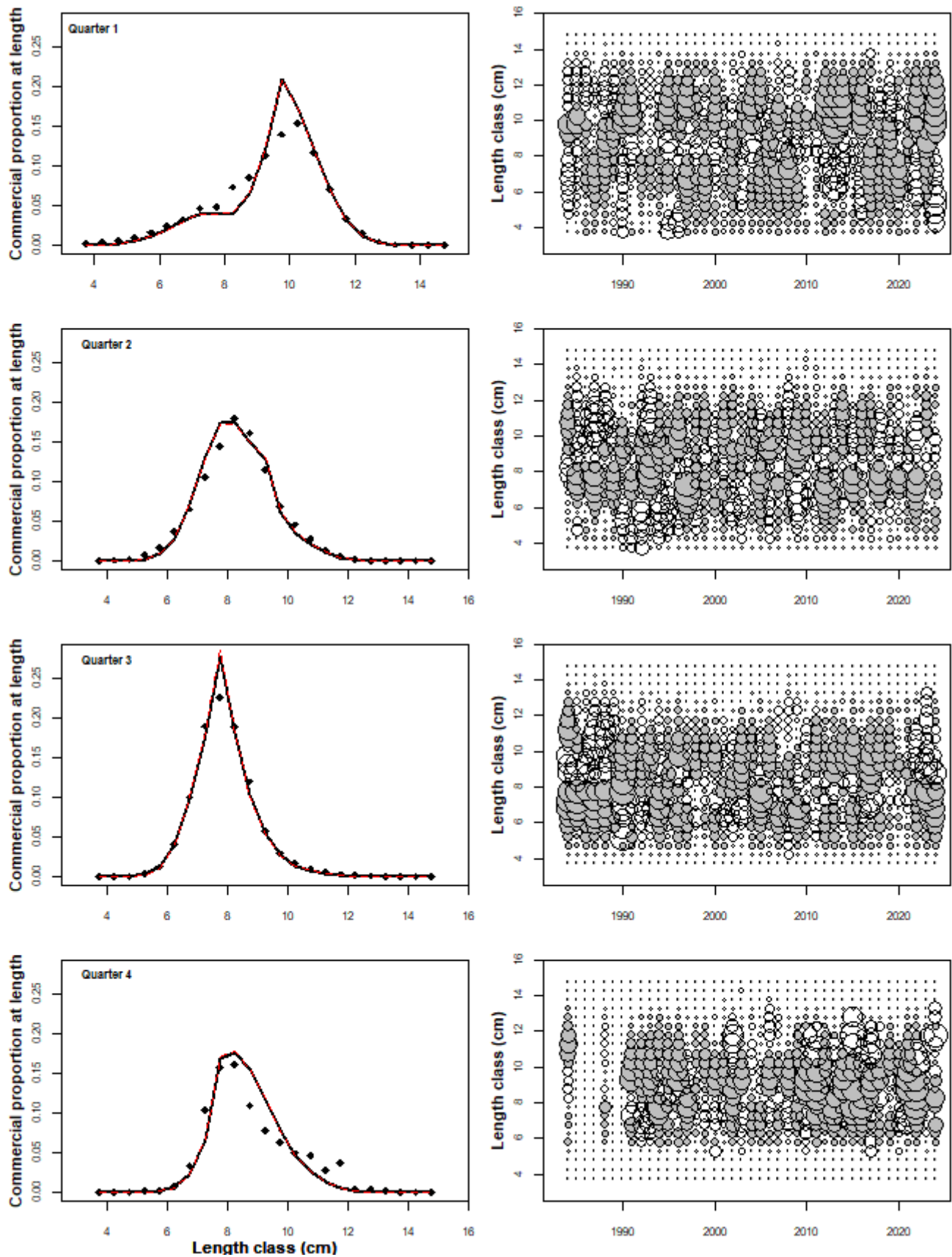


Figure 7. Average (over all years) model predicted and observed proportions-at-length in the quarterly commercial catch for A_{BH} (black) and A_{3BH} (red). The standardised residuals for A_{BH} are given in the bubble plot; the size of the bubbles are proportional to the absolute value of the residuals, while the shaded bubbles show negative and the unshaded bubbles show positive residuals.

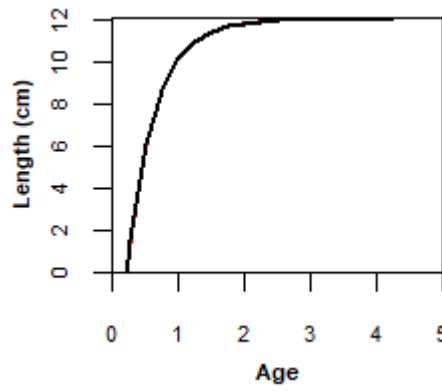


Figure 8. The model estimated von Bertalanffy growth curve, where integer ages are taken to correspond to 1 October for A_{BH} .

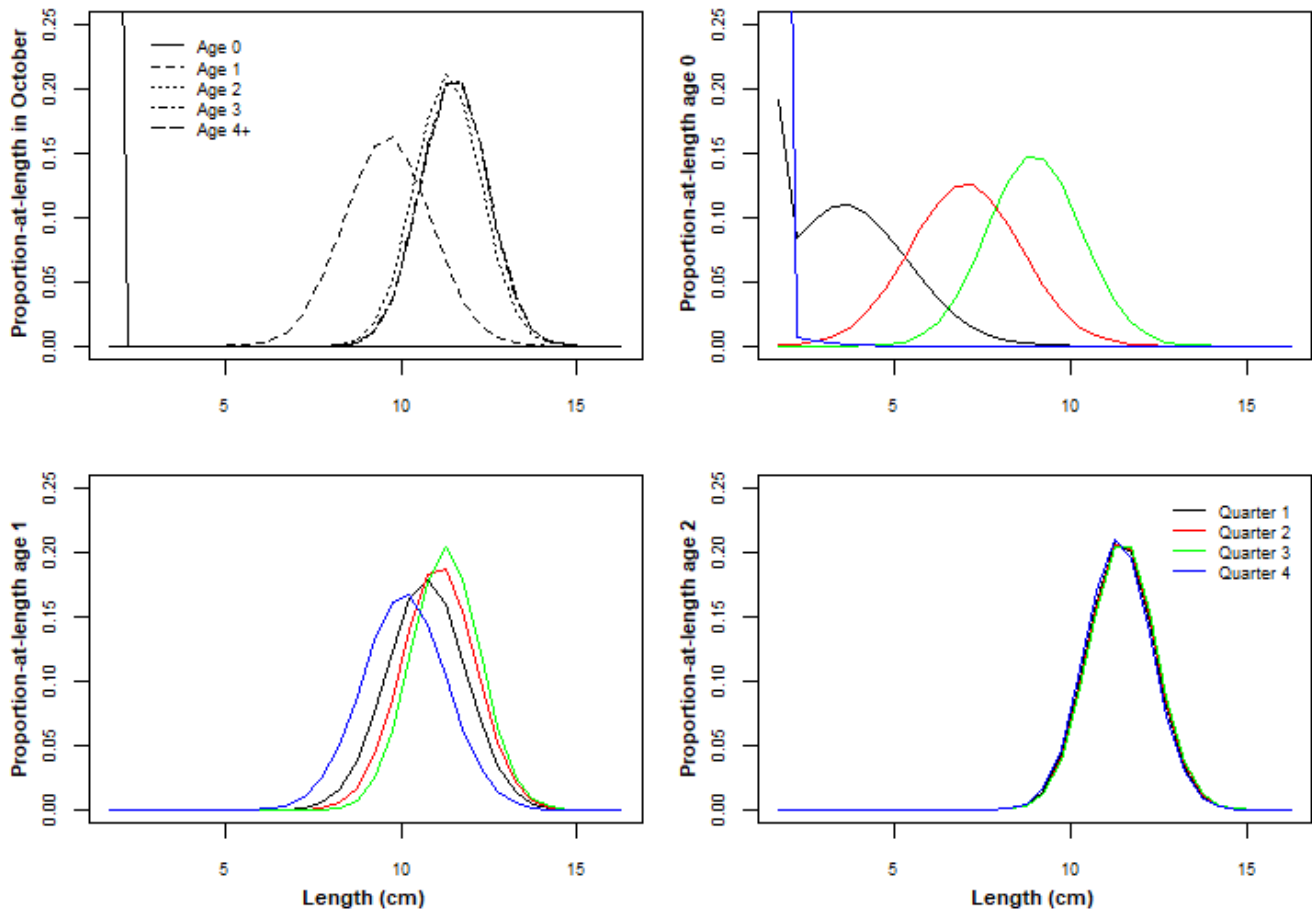


Figure 9. The model estimated distributions of proportions-at-length for each ages 0, 1 and 2, given at the middle of each quarter of the year (corresponding to the times commercial catch is modelled to be taken) and the distributions for all ages at 1 October (corresponding to the time of spawning) for A_{BH} .

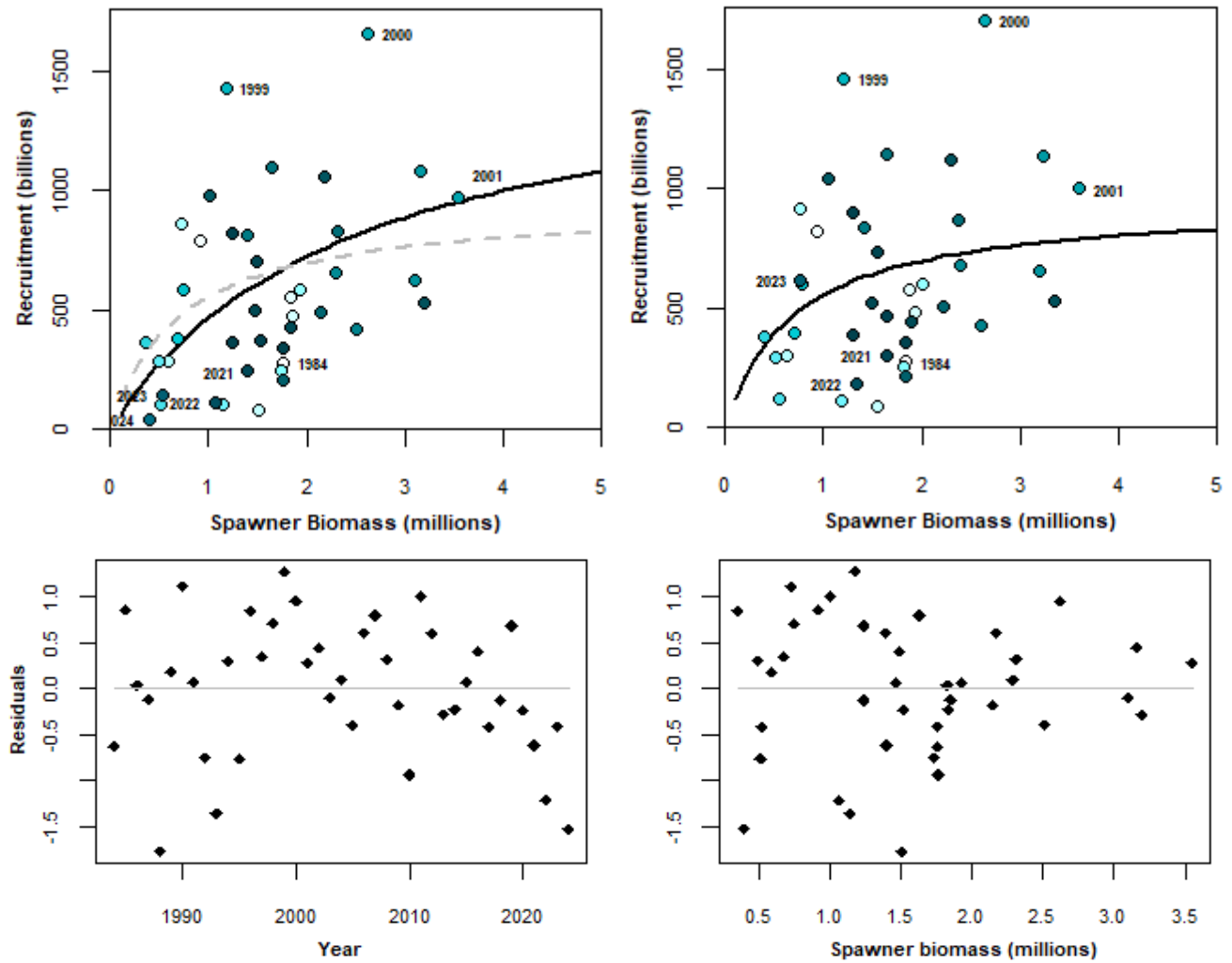


Figure 10a. Model predicted anchovy recruitment (at 1 October) plotted against spawner biomass from October 1984 to 2024 with the Beverton Holt stock recruitment relationship for A_{BH} (top left). The top right plot shows the relationship estimated by de Moor (2024) and this relationship is shown in grey in the top left plot. The residuals from the A_{BH} fit are given in the lower plots, against year and against spawner biomass. Note that the final year recruitment may not be well estimated.

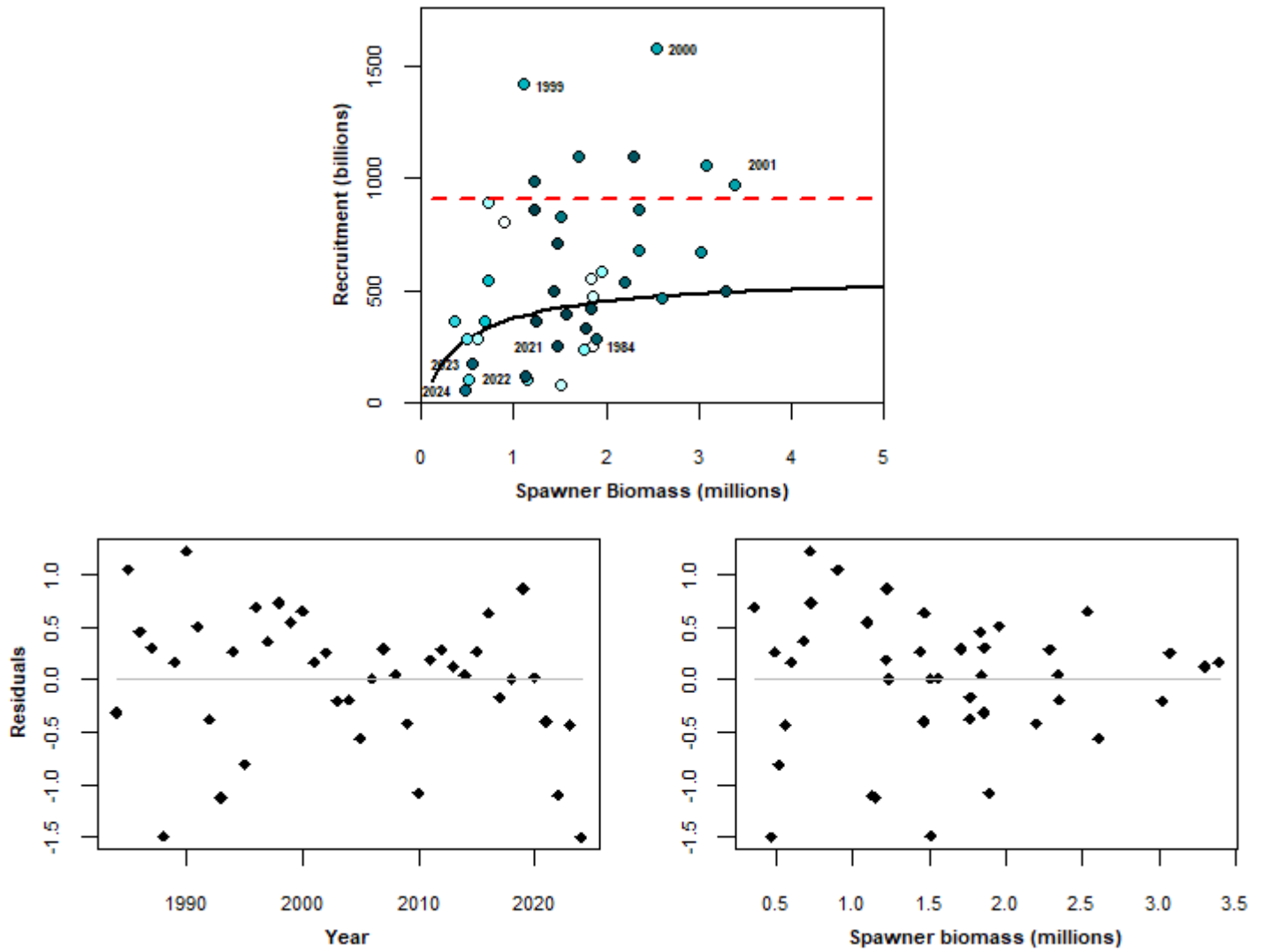


Figure 10b. Model predicted anchovy recruitment (at 1 October) plotted against spawner biomass from October 1984 to 2024 with the Beverton Holt stock recruitment relationship for A_{3BH} (top). The residuals are given in the lower plots, against year and against spawner biomass. The standard deviation for the early and later years ($\sigma_r = 0.70$) is greater than that for the middle years ($\sigma_r = 0.46$). Note that the final year recruitment may not be well estimated.

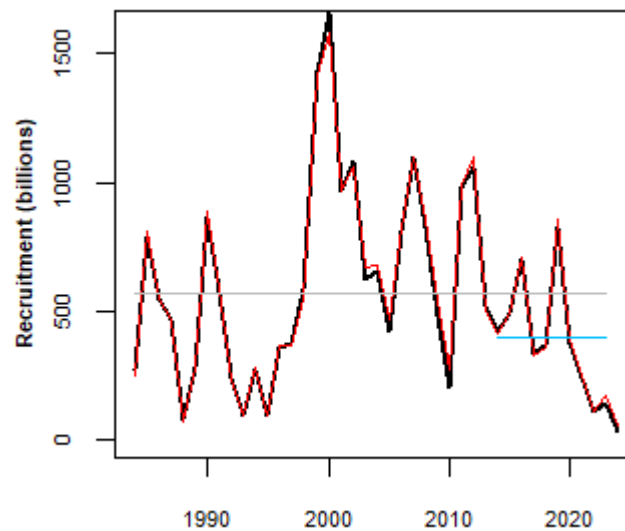


Figure 11. The timeseries of November recruitments for A_{BH} (black line) and A_{3BH} (red line). The grey and blue lines indicate the A_{BH} average over the full timeseries (570 billion, decreased from 610 billion of de Moor (2024)) and the most recent 10 years (406 billion, decreased from 481 billion of de Moor (2024)), respectively, excluding November 2024.

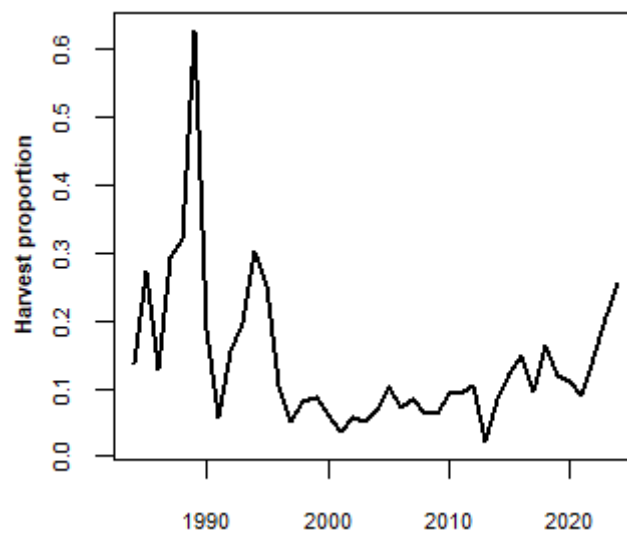


Figure 12. The historical harvest proportion estimated for A_{BH} .

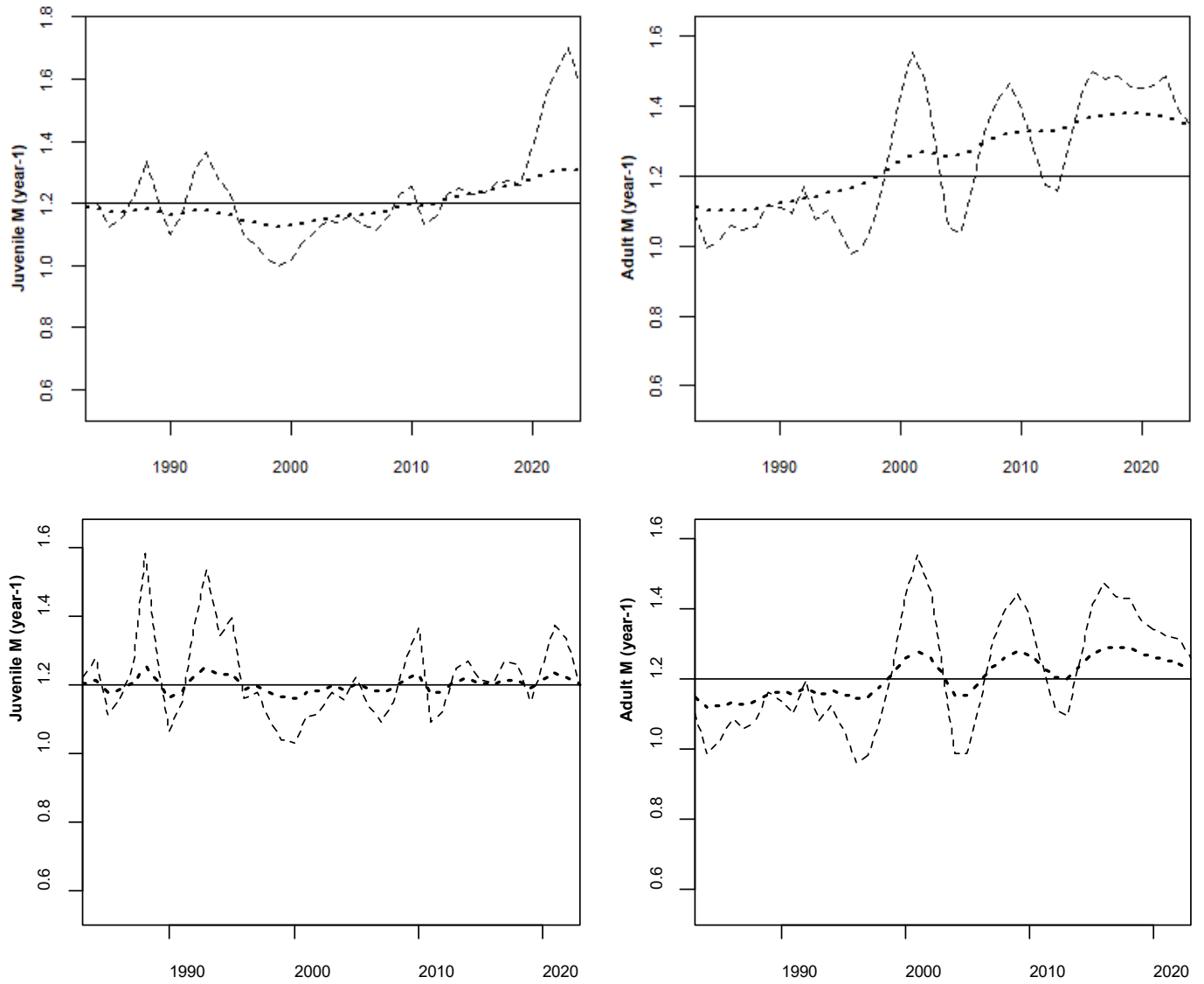


Figure 13. The estimated annual juvenile (left) and adult (right) natural mortality rates for A_{Mj} with $\sigma_j = 0.2$ (dashed) or $\sigma_j = 0.1$ (dotted) and A_{Mad} with $\sigma_{ad} = 0.2$ (dashed) or $\sigma_{ad} = 0.1$ (dotted). The equivalent figures from de Moor (2024) are shown in the lower plot for comparison.

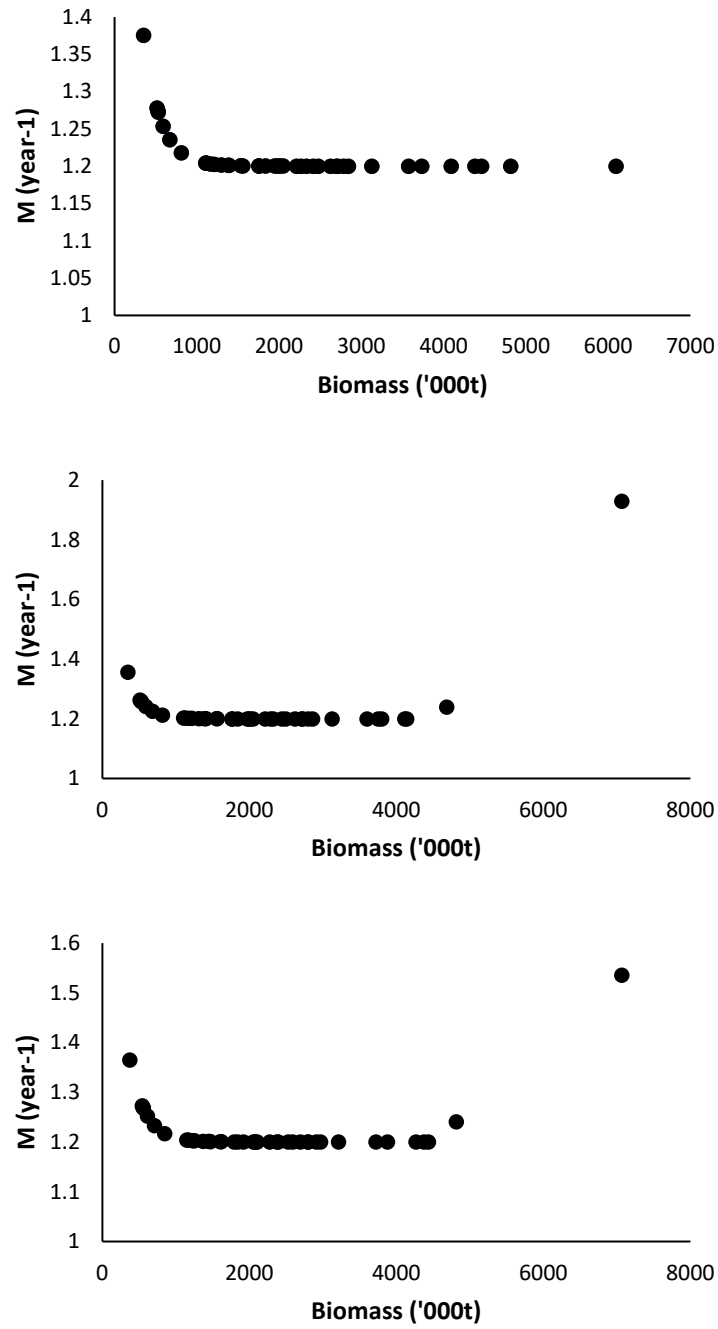


Figure 14. The estimated density dependent relationship between natural mortality and model estimated biomass A_{DDM1} ($\chi = 0.0050$, top row), A_{DDM2} ($\chi = 0.0054$, $\chi_2 = 0.0002$, 2nd row) and A_{DDM3} ($\chi = 0.0048$, $\chi_2 = 0.0001$, 3rd row).

Appendix A: Anchovy data for 2024

The following data are available for 2024 (Merkle *et al.* 2024, Coetzee *et al.* 2025, Merkle pers comm):

- May 2024 recruitment survey began at 3rd June, i.e. $t_y = 1.067$.
- May 2024 recruit cut-off length was <8.5cm.
- May 2024 recruit numbers and inter-transect CV for the area west of Cape Infanta was 54.229 billion (0.304).
- November 2024 total biomass survey took place over March-April 2025 and is modelled to have occurred on 1 April 2025 instead of 15th November 2024.
- The total biomass and inter-transect CV for the area west of Port Alfred was 116 197t (0.409).
- The March-April 2025 survey length frequencies.
- Commercial catch tonnages and length frequencies from January 2024 to March 2025.
- Catch from 1-2 June 2025 (to contribute to recruit catch from 1 April to the day before the survey).
- As maturity data could not be collected during November 2024, the average 2017-2020,2022 $L_{50,y}$ and δ_y^{mat} were used for November 2024, as assumed for November 2023 (Table B.3).

Appendix B: Bayesian operating model for the South African anchovy resource

In the below equations a “ $\hat{}$ ” is used to represent an estimate of a quantity (e.g. biomass) from a source external to this model (e.g. a survey). Model predicted quantities are represented by terms without any additional super-/sub-scripts other than dependencies on, for example, year, length etc.

Model Assumptions

- 1) Annual quarters are 1 January – 31 March, 1 April – 30 June, 1 July – 30 September and 1 October – 31 December.
- 2) Spawning and recruitment are assumed to occur at the start of the 4th quarter (1 October).
- 3) Anchovy mature according to a length-based ogive with an annually varying L_{50} .
- 4) A plus group of age 4 is used.
- 5) Two hydro-acoustic surveys are held each year: the first takes place in November and provides an index of abundance of the total biomass; the second is in May/June (known as the recruit survey) and provides an index of abundance of recruitment.
- 6) Both surveys provide relative indices of abundance.
- 7) The November survey is assumed to provide an index of abundance corresponding to 15th November (Figure A.1 of de Moor 2022). The 4th quarter catch is modelled to be taken in a pulse immediately before this estimate of biomass is calculated.
- 8) The recruit survey estimate of recruit abundance is assumed to correspond with recruitment on the day on which the survey began.
- 9) The egg survey observations (derived from data collected during the earlier November surveys) provide estimates of spawner biomass in absolute terms.
- 10) Pulse fishing occurs four times a year, in the middle of each quarter.

Population Dynamics

The assessment is begun at the beginning of the 4th quarter of $y_0 = 1983$, but conditioned to data from $y_1 = 1984$ to $y_n = 2024$.

The following subscript notation is used:

- quarters $q = 1$ denoting January to March, $q = 2$ denoting April to June, $q = 3$ denoting July to September and $q = 4$ denoting October to December;
- ages $a = 0$ to a plus group of $a = 4^+$;
- lengths from a minus group of $l = 1.5^-$ cm to a plus group of $l = 16^+$ cm, with length classes of size 0.5cm;

Numbers-at-age 1+ at 1 October

$$N_{y,a} = \left(\left(\left(\left(N_{y-1,a-1} e^{-M_{a-1,y/8}} - C_{y-1,4,a-1} \right) e^{-M_{a-1,y/4}} - C_{y,1,a-1} \right) e^{-M_{a-1,y/4}} - C_{y,2,a-1} \right) e^{-M_{a-1,y/4}} - C_{y,3,a-1} \right) e^{-M_{a-1,y/8}}$$

$$y_1 \leq y \leq y_n, 1 \leq a \leq 3$$

$$N_{y,4+} = \left(\left(\left((N_{y-1,3} e^{-M_{3,y/8}} - C_{y-1,4,3}) e^{-M_{3,y/4}} - C_{y,1,3} \right) e^{-M_{3,y/4}} - C_{y,2,3} \right) e^{-M_{3,y/4}} - C_{y,3,3} \right) e^{-M_{3,y/8}} \\ + \left(\left(\left((N_{y-1,4+} e^{-M_{4+,y/8}} - C_{y-1,4,4+}) e^{-M_{4+,y/4}} - C_{y,1,4+} \right) e^{-M_{4+,y/4}} - C_{y,2,4+} \right) e^{-M_{4+,y/4}} - C_{y,3,4+} \right) e^{-M_{4+,y/8}} \\ y_1 \leq y \leq y_n \text{ (AError! Bookmark not defined..1)}^2$$

Natural mortality

$$M_{0,y} = \bar{M}_j e^{\varepsilon_{j,y}} \text{ with } \varepsilon_{1984}^j = \eta_{1984}^j \text{ and } \varepsilon_y^j = \rho \varepsilon_{y-1}^j + \eta_y^j \sqrt{1 - \rho^2} \quad y_1 \leq y \leq y_{n+1} \text{ (B.2)}$$

$$M_{1+,y} = \bar{M}_{ad} e^{\varepsilon_{ad,y}} \text{ with } \varepsilon_{1984}^{ad} = \eta_{1984}^{ad} \text{ and } \varepsilon_y^{ad} = \rho \varepsilon_{y-1}^{ad} + \eta_y^{ad} \sqrt{1 - \rho^2} \quad y_1 \leq y \leq y_{n+1} \text{ (B.3)}$$

Numbers-at-length

The model predicted numbers-at-length of ages 1+ only at 1st October are given by:

$$N_{y,l}^{1+} = \sum_{a=1}^{4+} A_{a,l}^{Oct} N_{y,a} \quad y_1 \leq y \leq y_n, 1.5^- \text{ cm} \leq l \leq 16^+ \text{ cm} \text{ (B.4)}$$

Numbers-at-length at 15th November, after catch

$$N_{y,l} = \begin{cases} \sum_{a=0}^{4+} A_{a,l} (N_{y,a} e^{-M_{a,y+1/8}} - C_{y,4,a}) & y_1 \leq y \leq y_{n-2} \\ \sum_{a=0}^{4+} A_{1,a,l} [(N_{y,a} e^{-M_{a,y+1/8}} - C_{y,4,a}) e^{-7M_{a,y+1/24}}] & y = y_{n-1} \\ \sum_{a=0}^{4+} A_{1,a,l} [(N_{y,a} e^{-M_{a,y+1/8}} - C_{y,4,a}) e^{-3M_{a,y+1/8}}] & y = y_n \end{cases} \quad 1.5^- \text{ cm} \leq l \leq 16^+ \text{ cm} \text{ (B.5)}^3$$

Growth

The proportion of anchovy of age a that fall in the length group l is calculated under the assumption that length-at-age is normally distributed about a Richards or von Bertalanffy (if $D = 0$) growth curve. This is calculated at times during the year modelled to correspond to spawning (1 October) and commercial catches (midpoint of each quarter):

$$A_{a,l}^{Oct} \sim N \left(L_{\infty} (1 + (D - 1) e^{-\kappa(a-t_0)})^{1/(1-D)}, \vartheta_a^2 \right) \quad 0 \leq a \leq 4^+, 1.5^- \text{ cm} \leq l \leq 16^+ \text{ cm} \text{ (B.6)}^4$$

$$A_{q,a,l} \sim \begin{cases} N \left(L_{\infty} (1 + (D - 1) e^{-\kappa(a+(2q+1)/8-t_0)})^{1/(1-D)}, \left(1 - \frac{2q+1}{8} \right) \vartheta_a^2 + \frac{2q+1}{8} \vartheta_{a+1}^2 \right) & 1 \leq q \leq 3 \\ N \left(L_{\infty} (1 + (D - 1) e^{-\kappa(a+1/8-t_0)})^{1/(1-D)}, \left(1 - \frac{1}{8} \right) \vartheta_a^2 + \frac{1}{8} \vartheta_{a+1}^2 \right) & q = 4 \end{cases} \quad 0 \leq a \leq 4^+, 1.5^- \text{ cm} \leq l \leq 16^+ \text{ cm} \text{ (B.7)}^7$$

Spawning biomass at 1 October

$$SSB_y = \sum_{l=1.5^-}^{16^+} f_{y,l} N_{y,l}^{1+} w_l \quad y_1 \leq y \leq y_n \text{ (B.8)}$$

where

$$f_{y,l} = 1 / \left(1 + e^{-(l-L_{50,y})/\delta_y^{mat}} \right) \quad y_1 \leq y \leq y_n \text{ (B.9)}^5$$

² $C_{1983,4,a} = 0$ as $C_{1983,m,l}^{RLF}$ is unknown.

³ The adjustments for $y_{n-1} = 2023$ and $y_n = 2024$ is to 1st March 2024 and 1st April 2025, respectively. This for the delayed November 2023 survey which took place from mid-February to mid-March 2024 and for the delayed November 2024 survey which took place from mid-March to mid-April 2025. Anchovy catch landed between January and March in these years was small and is thus ignored.

⁴ The proportion is calculated as the area under the curve between the lower limit and upper limit of length class l . The lower and upper tails are included in the proportions calculated for the minus and plus groups, respectively.

⁵ Where length is used in an equation, the mid-point of the length class is used.

Recruitment at 1 October

$$N_{y,0} = \frac{\alpha SSB_y}{\beta + SSB_y} e^{\varepsilon_y - 0.5\sigma_r^2} \quad y_1 \leq y \leq y_n \quad (\text{B.10})$$

Biomass associated with the November survey

$$B_y = k_N \sum_{l=1.5^-}^{16^+} N_{y,l} w_l \quad y_1 \leq y \leq y_n \quad (\text{B.11})$$

Proportion-at-length associated with the November survey

$$p_{y,l} = \frac{N_{y,l} S_l^{sur}}{\sum_{l=2}^{15.5} N_{y,l} S_l^{sur}} \quad y_1 \leq y \leq y_n, \quad (\text{B.12})$$

where

$$S_l^{sur} = \begin{cases} 0 & l = 1.5^- \\ \frac{1}{1 + \exp(-(l - l^{sur})/\delta^{sur})} & 2cm \leq l \leq 15.5cm \\ 0 & l = 16^+ \end{cases} \quad (\text{B.13})^6$$

Commercial selectivity

Commercial selectivity-at-length is assumed to follow the logistic shape, with a dome at high lengths. Commercial selectivity is assumed to vary by quarter, but remain unchanged over time. Selectivity-at-lengths less than the smallest observed length class (3.5cm) and greater than the largest observed length class (14.5cm) are taken to be zero:

$$S_{y,q,l} = \begin{cases} 0 & 1.5^-cm \leq l \leq 3cm \\ 1/(1 + e^{\psi_q(l - l_{50q})}) & 3.5cm \leq l \leq S_q^{break} \\ S_{y,q,l-1} e^{\delta_q} & S_q^{break} \leq l \leq 14.5cm \\ 0 & 15cm \leq l \leq 16^+cm \end{cases} \quad y_1 \leq y \leq y_n, 1 \leq q \leq 4 \quad (\text{B.14})^7$$

Commercial selectivity-at-age is given by:

$$S_{y,q,a} = \sum_{l=1.5^-}^{16^+} A_{q,a,l} S_{y,q,l} \quad y_1 \leq y \leq y_n, 1 \leq q \leq 4, 0 \leq a \leq 4^+ \quad (\text{B.15})$$

Commercial catch

$$C_{y,4,a} = N_{y,a} e^{-M_{a,y+1/8}} S_{y,4,a} F_{y,4}$$

$$C_{y,1,a} = (N_{y-1,a} e^{-M_{a,y/8}} - C_{y-1,4,a}) e^{-M_{a,y/4}} S_{y,1,a} F_{y,1}$$

$$C_{y,2,a} = ((N_{y-1,a} e^{-M_{a,y/8}} - C_{y-1,4,a}) e^{-M_{a,y/4}} - C_{y,1,a}) e^{-M_{a,y/4}} S_{y,2,a} F_{y,2}$$

$$C_{y,3,a} = (((N_{y-1,a} e^{-M_{a,y/8}} - C_{y-1,4,a}) e^{-M_{a,y/4}} - C_{y,1,a}) e^{-M_{a,y/4}} - C_{y,2,a}) e^{-M_{a,y/4}} S_{y,3,a} F_{y,3}$$

$$y_1 \leq y \leq y_n, 0 \leq a \leq 4^+ \quad (\text{B.16})$$

⁶ A zero survey trawl selectivity is assumed for lengths 1.5cm and 16cm, consistent with the range of length classes in the observed trawl survey proportions-at-length.

⁷ These selectivities-at-length are renormalized so that the maximum is 1.

The fished proportion of the available biomass from the anchovy fishery is estimated by:

$$F_{y,4} = \frac{\sum_{m=10}^{12} \sum_{l=3.5}^{14.5} C_{y,m,l}^{RLF}}{\sum_{a=0}^{4+} N_{y,a} e^{-M_{a,y+1/8}} S_{y,4,a}}$$

$$F_{y,1} = \frac{\sum_{m=1}^3 \sum_{l=3.5}^{14.5} C_{y,m,l}^{RLF}}{\sum_{a=0}^{4+} (N_{y-1,a} e^{-M_{a,y/8}} - C_{y-1,4,a}) e^{-M_{a,y/4}} S_{y,1,a}}$$

$$F_{y,2} = \frac{\sum_{m=4}^6 \sum_{l=3.5}^{14.5} C_{y,m,l}^{RLF}}{\sum_{a=0}^{4+} ((N_{y-1,a} e^{-M_{a,y/8}} - C_{y-1,4,a}) e^{-M_{a,y/4}} - C_{y,1,a}) e^{-M_{a,y/4}} S_{y,2,a}}$$

$$F_{y,3} = \frac{\sum_{m=7}^9 \sum_{l=3.5}^{14.5} C_{y,m,l}^{RLF}}{\sum_{a=0}^{4+} (((N_{y-1,a} e^{-M_{a,y/8}} - C_{y-1,4,a}) e^{-M_{a,y/4}} - C_{y,1,a}) e^{-M_{a,y/4}} - C_{y,2,a}) e^{-M_{a,y/4}} S_{y,3,a}} \quad y_1 \leq y \leq y_n \text{ (B.17)}^8$$

A penalty is imposed within the model to ensure that $S_{y,q,l} F_{y,q} < 0.95$ for all l .

Number of recruits at the time of the recruit survey

$$N_{y,r} = k_r \left((N_{y-1,0} e^{-M_{0,y/8}} - C_{y-1,4,0}) e^{-M_{0,y/4}} - C_{y,1,0} \right) e^{-(1/8+0.5(1+t_y)/12)M_{0,y}} - C_{y,0bs} e^{-(0.5(1+t_y)/12)M_{0,y}} \quad y_2 \leq y \leq y_n \text{ (B.18)}$$

The recruit catch from 1 April to the day before the survey is calculated as follows

$$C_{y,0bs} = ((N_{y-1,0} e^{-M_{0,y/8}} - C_{y-1,4,0}) e^{-M_{0,y/4}} - C_{y,1,0}) e^{-(1/8+0.5(1+t_y)/12)M_{0,y}} S_{y,2,0} F_{y,bs} \quad y_2 \leq y \leq y_n \text{ (B.19)}$$

where

$$F_{y,bs} = \frac{\sum_{l=3.5}^{14.5} C_{y,bs,l}^{RLF}}{\sum_{a=0}^{4+} ((N_{y-1,a} e^{-M_{a,y/8}} - C_{y-1,4,a}) e^{-M_{a,y/4}} - C_{y,1,a}) e^{-(1/8+0.5(1+t_y)/12)M_{a,y}} S_{y,2,a}} \quad y_2 \leq y \leq y_n \text{ (B.20)}$$

A penalty is imposed within the model to ensure that $S_{y,2,l} F_{y,bs} < 0.95$ for all l .

Commercial catch-at-length from the anchovy fishery

$$C_{y,4,l} = \sum_{a=0}^{4+} N_{y,a} e^{-M_{a,y+1/8}} A_{4,a,l} S_{y,4,l} F_{y,4}$$

$$C_{y,1,l} = \sum_{a=0}^{4+} (N_{y-1,a} e^{-M_{a,y/8}} - C_{y-1,4,a}) e^{-M_{a,y/4}} A_{1,a,l} S_{y,1,l} F_{y,1}$$

$$C_{y,2,l} = \sum_{a=0}^{4+} ((N_{y-1,a} e^{-M_{a,y/8}} - C_{y-1,4,a}) e^{-M_{a,y/4}} - C_{y,1,a}) e^{-M_{a,y/4}} A_{2,a,l} S_{y,2,l} F_{y,2}$$

$$C_{y,3,l} = \sum_{a=0}^{4+} (((N_{y-1,a} e^{-M_{a,y/8}} - C_{y-1,4,a}) e^{-M_{a,y/4}} - C_{y,1,a}) e^{-M_{a,y/4}} - C_{y,2,a}) e^{-M_{a,y/4}} A_{3,a,l} S_{y,3,l} F_{y,3} \quad y_1 \leq y \leq y_n, 1.5^- \text{ cm} \leq l \leq 16^+ \text{ cm} \text{ (B.21)}$$

Proportion-at-length associated with the commercial catch

$$p_{y,q,l}^{coml} = \frac{C_{y,q,l}}{\sum_{l=3.5}^{14.5} C_{y,q,l}} \quad y_1 \leq y \leq y_n, 1 \leq q \leq 4, 3.5 \text{ cm} \leq l \leq 14.5 \text{ cm} \text{ (B.22)}^9$$

Fitting the Model to Observed Data (Likelihood)

The survey observations of abundance are assumed to be log-normally distributed. The standard errors of the log-distributions for the survey observations of adult biomass and recruitment numbers are approximated by the CVs of the untransformed

⁸ The range of length classes used in these summation matches the range of length classes in the observations which is a smaller range than the maximum range modelled of 1.5 cm to 16⁺ cm.

⁹ Note the model predicted commercial catch of lengths <3.5 cm and >14.5 cm are zero (equation A.14), consistent with the range of length classes observed in the commercial catch.

distributions and a further additional variance parameter. A “sqrt(p)” formulation, rather than the “adjusted lognormal” (“Punt-Kennedy”, Punt and Kennedy 1997) error distribution formulation, is assumed for the estimated proportions-at-length particularly as it can deal with occasional zero observations more easily. This “sqrt(p)” formulation mimics a multinomial form for the error distribution by forcing near-equivalent variance-mean relationship for the error distributions. The negative log-likelihood function is given by:

$$-lnL = -lnL^{Nov} - lnL^{Egg} - lnL^{rec} - lnL^{surpropl} - lnL^{compropl} \quad (B.23)$$

where

$$-lnL^{Nov} = \frac{1}{2} \sum_{y=y_1}^{y_n} \left\{ \frac{(\ln \hat{B}_y - \ln(B_y))^2}{\sigma_{y,N}^2 + \phi_{ac}^2 + \lambda_N^2} + \ln(2\pi(\sigma_{y,N}^2 + \phi_{ac}^2 + \lambda_N^2)) \right\} \quad (B.24)^{10}$$

$$-lnL^{Egg} = \frac{1}{2} \sum_{y=y_1}^{1993} \left\{ \frac{(\ln \hat{B}_{y,egg} - \ln(k_g SSB_y))^2}{\sigma_{y,egg}^2} + \ln(2\pi\sigma_{y,egg}^2) \right\} \quad (B.25)$$

$$-lnL^{rec} = \frac{1}{2} \sum_{y=y_1+1}^{y_n} \left\{ \frac{(\ln \hat{N}_{y,r} - \ln(N_{y,r}))^2}{\sigma_{y,r}^2 + \phi_{ac}^2 + \lambda_r^2} + \ln(2\pi(\sigma_{y,r}^2 + \phi_{ac}^2 + \lambda_r^2)) \right\} \quad (B.26)^{11}$$

$$-lnL^{surpropl} = w_{propl}^{sur} \sum_{y=y_1}^{y_n} \sum_{l=2}^{15.5} \left\{ \frac{(\sqrt{\hat{p}_{y,l}} - \sqrt{p_{y,l}})^2}{2\sigma_{sur}^2} + \ln(\sigma_{sur}) \right\}^{12} \quad (B.27)$$

$$-lnL^{compropl} = w_{propl}^{com} \sum_{y=y_1}^{y_n} \sum_{q=1}^4 \sum_{l=3.5}^{14.5} \left\{ \frac{(\sqrt{\hat{p}_{y,q,l}^{coml}} - \sqrt{p_{y,q,l}^{coml}})^2}{2\sigma_{com}^2} + \ln(\sigma_{com}) \right\} \quad (B.28)$$

¹⁰ Excluding 2021 when no survey was completed.

¹¹ Excluding 2018 when no survey was completed.

¹² Although strictly there may be bias in the proportions of length-at-age data, no bias is assumed in this assessment. The effect of such a bias is assumed to be small.

Table B.1. Assessment model parameters and variables.

Parameter / Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation	Notes
$N_{y,a}$	Model predicted numbers-at-age a at 1 October in year y	Billions		B.1 & B.10	
Annual numbers and biomass	$N_{1983,a}$	Billions	$N_{1983,0} \sim N(53, 100^2),$ $N_{1983,1} \sim N(482, 200^2),$ $N_{1983,2} \sim N(1, 2^2),$ $N_{1983,3} = N_{1983,2} e^{-M_{2,1983}^A},$ $N_{1983,4} = N_{1983,3} \times \frac{e^{-M_{3,1983}^A}}{1 - e^{-M_{3,1983}^A}}$		Hardly informative ¹³ $M_{ad,1983}^A = M_{ad,1984}^A$
	$N_{y,l}$	Billions		B.5	
	$N_{y,l}^{1+}$	Billions		B.4	
	B_y	Thousand tons		B.11	
	w_l	Grams	$w_l = 0.007366 \times l^{3.11124}$		de Moor (2023)
	SSB_y	Thousand tons		B.8	
	$f_{y,l}$			B.9	
	$L_{50,y}$	Cm	Table B.3		Yemane <i>et al.</i> 2024
	δ_y^{mat}	-	Table B.3		Yemane <i>et al.</i> 2024
Catch	$C_{y,q,a}$	Billions		B.16	
	$F_{y,q}$	-		B.17	
	$C_{y,obs}$	Billions		B.19	
	$F_{y,bc}$	-		B.20	

¹³ Based on difficulties with convergence to the posterior distribution from previous assessments (de Moor 2016), “hardly informative” prior distributions have been used for some parameters which do no more than simply aid the software to conduct MCMC simulations. Normal prior distributions were assigned to these parameters with means roughly corresponding to the baseline parameter values giving the minimum objective function value. The standard deviations for these prior distributions were chosen such that the Hessian-based SE resulting from the model fit was less (as much less as possible) than that of the prior distribution.

¹⁴ Where length is required in an equation, the mid-point of the length class is used.

Table B.1 (continued).

Parameter / Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation	Notes
Natural Mortality	$M_{a,y}$	Rate of natural mortality of age a in year y	Year ⁻¹	B.2 & B.3	
	$\bar{M}_j$	Median rate of natural mortality for age-0 anchovy	Year ⁻¹	1.2	de Moor (2024)
	$\bar{M}_{ad}$	Median rate of natural mortality for 1+ anchovy	Year ⁻¹	1.2	
	ε_y^j	Annual residuals about natural mortality rate for age-0 anchovy	-	B.2	
	ε_y^{ad}	Annual residuals about natural mortality rate for 1+ anchovy	-	B.3	
	η_y^j	Normally distributed error in calculating ε_y^j	-	$\sim N(0, \sigma_j^2)$	
	η_y^{ad}	Normally distributed error in calculating ε_y^{ad}	-	$\sim N(0, \sigma_{ad}^2)$	
	σ_j	Standard deviation in the annual residuals η_y^j	-	0.1 or 0.2	de Moor (2024) and this document
	σ_{ad}	Standard deviation in the annual residuals η_y^{ad}	-	0.1 or 0.2	
	ρ	Annual autocorrelation coefficient	-	$\sim U(0,1)$	
Selectivity	S_l^{sur}	November survey trawl selectivity-at-length l	-	B.13	
	l^{sur}	Length class number at which the survey selectivity-at-length is 50%	Cm	$U(2.5,11.5)$	
	δ^{sur}	Steepness of the survey selectivity-at-length relationship		$U(0.005,5)$	
	$S_{y,q,l}$	Commercial selectivity-at-length l during quarter q of year y	-	B.14	
	$S_{y,q,a}$	Commercial selectivity-at-age a during quarter q of year y	-	B.15	
	ψ_q	Steepness of ascending limb of logistic part of commercial selectivity curve during quarter q	-	$\sim U(-10,0), \psi_1 = \psi_2 = \psi_3$	Uninformative
	$l50_q$	Length at which ascending limb of logistic part of commercial selectivity is 50% during quarter q	Cm	$\sim U(3,10), l50_3 = l50_4$	Uninformative
	δ_q	Rate of exponential decrease in commercial selectivity at large lengths during quarter q	-	$\delta_1 \sim N(-0.49, 0.5^2),$ $\delta_2 \sim N(-0.75, 1.0^2),$ $\delta_3 = \delta_4 \sim N(-0.66, 0.2^2)$	Hardly informative ¹⁶
	S_q^{break}	Length at which commercial selectivity starts to decrease during quarter q	Cm	$S_1^{break} = 9.5, S_2^{break} = 9,$ $S_3^{break} = S_4^{break} = 7.5$	

Table B.1 (Continued).

Parameter/ Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation	Notes
h	Steepness associated with the stock-recruitment curve ¹⁵	-	$h \sim U(0.2, 1)$		
K	Carrying capacity	Thousand tons	$K/1000 \sim U(0, 10)$		
α	Stock-recruitment curve parameter, related to h and K , for Beverton Holt and Ricker curves	-	$= \frac{4hK}{(5h-1) \left(\sum_{a=1}^3 \bar{f}_a w_a e^{-\bar{M}_j - (a-1)\bar{M}_{ad}} + \bar{f}_{4+} w_{4+} e^{-\bar{M}_j - 3\bar{M}_{ad}} / (1 - e^{-\bar{M}_{ad}}) \right)}$		
β	Stock-recruitment curve parameter, related to h and K , for Beverton Holt and Ricker curves	Thousand tons	$= \frac{(1-h)K}{(5h-1)}$		
ε_y	Annual lognormal deviation of recruitment	-	$\sim N(0, \sigma_r^2), y_1 \leq y \leq 1999$ $\sim N(0, \sigma_{r,2000+}^2), 2000 \leq y \leq y_n$		
σ_r^2	Variance in the residuals (lognormal deviation) about the stock recruitment curve	-	$\sim U(0.16, 10)$		Lower bound chosen to restrict the influence of the stock recruitment curve on the assessment results
$\bar{f}_a$	Average proportion of anchovy mature at age a		$= \frac{1}{y_n - y_1 + 1} \sum_{y=y_1}^{y_n} \sum_{l=1.5-}^{16+} f_{y,l} A_{a,l}^{Oct}$		
w_a	Mean mass at age a at spawning		$= \sum_{l=1.5-}^{16+} w_l A_{a,l}^{Oct}$		
$N_{y,r}$	Model predicted number of recruits at the time of the recruit survey in year y	Billions		B.18	

¹⁵ The proportion of the median virgin recruitment that is realised at a spawning biomass level of 20% of average pre-exploitation (virgin) spawning biomass, K .

Table B.1 (Continued).

Parameter / Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation	Notes
Multiplicative bias	k_N	-	$\ln(k_N) \sim N(-0.155, 0.391^2)$		de Moor <i>et al.</i> (2024a)
	k_g	-	1.0		From de Moor <i>et al.</i> (2020b)
	k_r	-	$k_r/k_N \sim U(0,1)$		Recruit survey assumed to cover less of the recruits than the November survey covers of the total biomass
Proportions-at-length and growth	$p_{y,l}$	-		B.12	
	$A_{a,l}^{Oct}$	-		B.6	
	$p_{y,q,l}^{com}$	-		B.22	
	$A_{q,a,l}$	-		B.7	
	L_∞	Cm	$\sim N(12, 1^2)$		Hardly informative ¹⁶
	κ	Year ⁻¹	$\frac{\kappa \times L_\infty}{10} \sim N(2.1, 1^2)$		Hardly informative ¹⁶
	t_0	Year	$\sim N(0.1, 1^2)$		Hardly informative ¹⁶
	ϑ_a	-	$\vartheta_0 \sim N(2, 1.5^2),$ $\vartheta_1 \sim N(1.3, 1.5^2),$ $\vartheta_{2+} \sim N(1.0, 0.1^2)$		Hardly informative ¹⁶

Table B.1 (Continued).

Parameter / Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation	Notes
$-\ln L^{Nov}$	Contribution to the negative log likelihood from the model fit to the November total survey biomass data	-		B.24	
$-\ln L^{Egg}$	Contribution to the negative log likelihood from the model fit to the egg survey spawner biomass data			B.25	
$-\ln L^{rec}$	Contribution to the negative log likelihood from the model fit to the recruit survey data	-		B.26	
$-\ln L^{surpropL}$	Contribution to the negative log likelihood from the model fit to the November survey proportion-at-length data	-		B.27	
$-\ln L^{compropL}$	Contribution to the negative log likelihood from the model fit to the quarterly commercial proportion-at-length data	-		B.28	
Likelihood	λ_N^2	-	0		See robustness tests
	λ_r^2	-	$\sim U(0,10)$		Uninformative
	ϕ_{ac}^2	-	0.197 ²		de Moor <i>et al.</i> (2024a)
	w_{prop}^{sur}	-	0.2		To allow for autocorrelation ¹⁶
	σ_{sur}	-	$\sqrt{\frac{\sum_{y=y_1}^{y_n} \sum_{l=7}^{13} (\sqrt{\hat{p}_{y,l}} - \sqrt{p_{y,l}})^2}{\sum_{y=y_1}^{y_n} \sum_{l=7}^{13} 1}}$		Closed form solution ¹⁷
	w_{prop}^{com}	-	0.05		To allow for autocorrelation ¹⁸
	σ_{com}	-	$\sqrt{\frac{\sum_{y=y_1}^{y_n} \sum_{q=1}^4 \sum_{l=5}^{12} (\sqrt{\hat{p}_{y,q,l}^{coml}} - \sqrt{p_{y,q,l}^{coml}})^2}{\sum_{y=y_1}^{y_n} \sum_{q=1}^4 \sum_{l=5}^{12} 1}}$		Closed form solution ¹⁹

¹⁶ Based upon data being available ~5 times more frequently than annual age data which contain maximum information content on this¹⁷ A shorter range of lengths ($7cm \leq l \leq 13cm$) is used given the near absence of data outside this range, resulting in small/zero residuals, which would negatively bias this estimate.¹⁸ Based upon data being available ~4x5 times more frequently than annual age data which contain maximum information content on this¹⁹ A shorter range of lengths ($5cm \leq l \leq 12cm$) is used given the near absence of data outside this range, resulting in small/zero residuals, which would negatively bias this estimate.

Table B.2. Assessment model data, detailed in de Moor *et al.* (2024b) and Appendix A.

Quantity	Description	Units / Scale	Shown in Figure
$C_{y,m,l}^{RLF}$	Observed number of anchovy in length class l caught during month m of year y . ²⁰	Billions	
$C_{y,bs}^{RLF}$	Observed number of anchovy in length class l caught from 1 April to the day before the start of the recruit survey in year y	Billions	
t_y	Time lapsed between 1 May and the start of the recruit survey in year y	Months	
$\hat{B}_y$	Acoustic survey estimate of total biomass from the November survey in year y	Thousand tons	Figure 1
$\sigma_{y,N}$	Survey sampling CV associated with $\hat{B}_y$ that reflects survey inter-transect variance	-	Figure 1
$\hat{B}_{y,egg}$	Egg survey estimate of spawner biomass from the November survey in year y	Thousand tons	Figure 2
$\sigma_{y,egg}$	Survey sampling CV associated with $\hat{B}_{y,egg}$ estimated from inter-transect variance		Figure 2
$\hat{N}_{y,r}$	Acoustic survey estimate of recruitment from the recruit survey in year y	Billions	Figure 3
$\sigma_{y,r}$	Survey sampling CV associated with $\hat{N}_{y,r}$ that reflects survey inter-transect variance	-	Figure 3
$\hat{p}_{y,l}$	Observed proportion (by number) of anchovy in length group l in the November survey of year y	-	
$\hat{p}_{y,q,l}^{com}$	Observed proportion (by number) of anchovy commercial catch in length group l during quarter q of year y		

Table B.3. Annual length (in cm) at which maturity is 50% and associated steepness (Yemane *et al.* 2024).

Year	$L_{50,y}$	δ_y^{mat}	Year	$L_{50,y}$	δ_y^{mat}	Year	$L_{50,y}$	δ_y^{mat}
1984 ²¹	8.891	1.250	1998	10.367	2.668	2012	8.393	1.550
1985	8.891	1.250	1999	9.557	2.678	2013	8.848	1.298
1986	8.495	1.304	2000	9.809	1.551	2014	10.034	1.962
1987	8.169	1.116	2001	10.179	1.833	2015	9.449	1.575
1988	7.987	1.366	2002	10.058	1.592	2016	9.012	0.848
1989	7.639	1.397	2003	9.673	1.414	2017	8.910	1.300
1990	7.793	0.868	2004	8.179	3.605	2018	8.890	1.734
1991	8.461	1.664	2005	9.069	0.359	2019	8.724	0.768
1992	8.887	1.695	2006	9.568	1.979	2020	9.380	2.830
1993	7.691	1.740	2007	9.922	1.730	2021 ²²	9.085	1.985
1994	7.528	1.839	2008	10.045	1.340	2022	8.789	1.141
1995	8.547	1.763	2009	10.101	2.327	2023 ²³	8.939	1.555
1996	9.406	2.034	2010	9.467	1.873	2024 ²⁷	8.939	1.555
1997	9.648	1.732	2011	8.292	2.906			

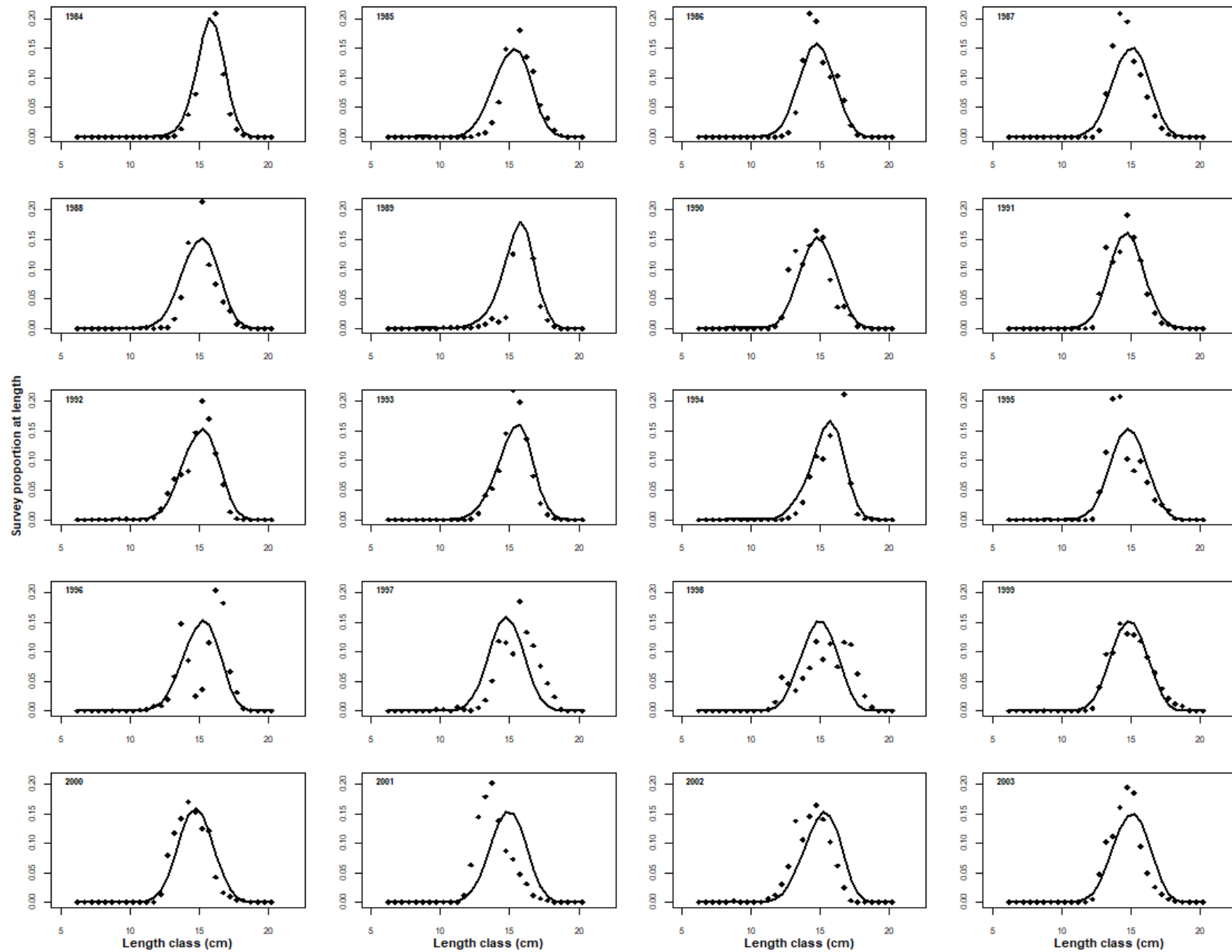
²⁰ This is the observed length-frequency adjusted such that the expected mass calculated using the weight-at-length relationship matches the observed catch in tons (de Moor *et al.* 2024b).

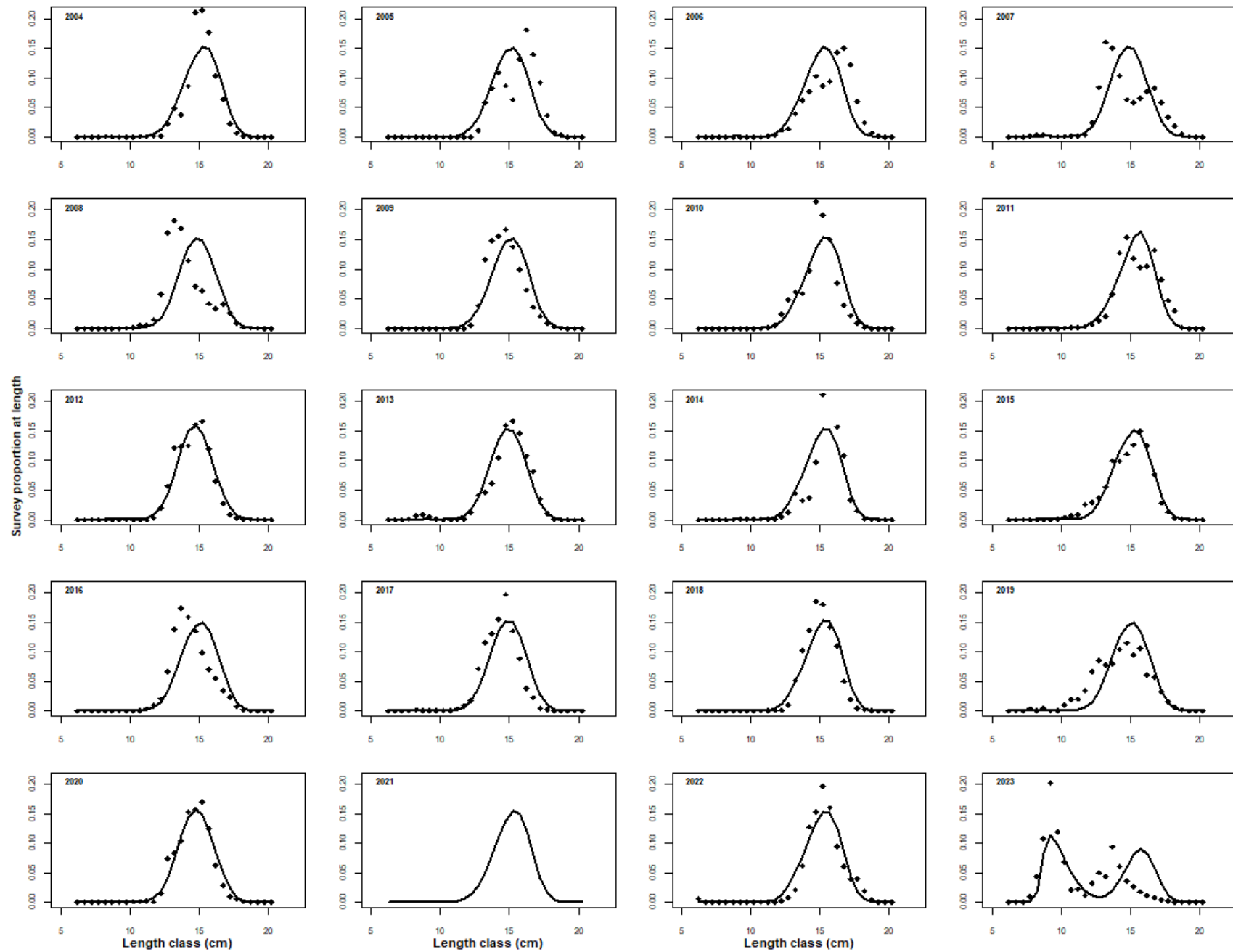
²¹ No data available. Set equal to 1985.

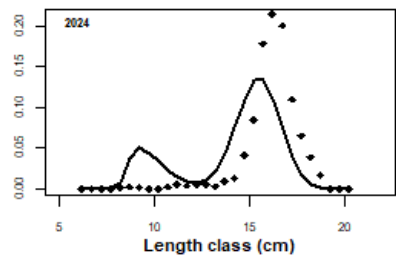
²² No data available. Used average 2020, 2022.

²³ No data for November 2023 and 2024 available. Used average 2017-2020, 2022.

Appendix C: Model predicted and observed proportions-at-length in the November surveys for A_{BH} .

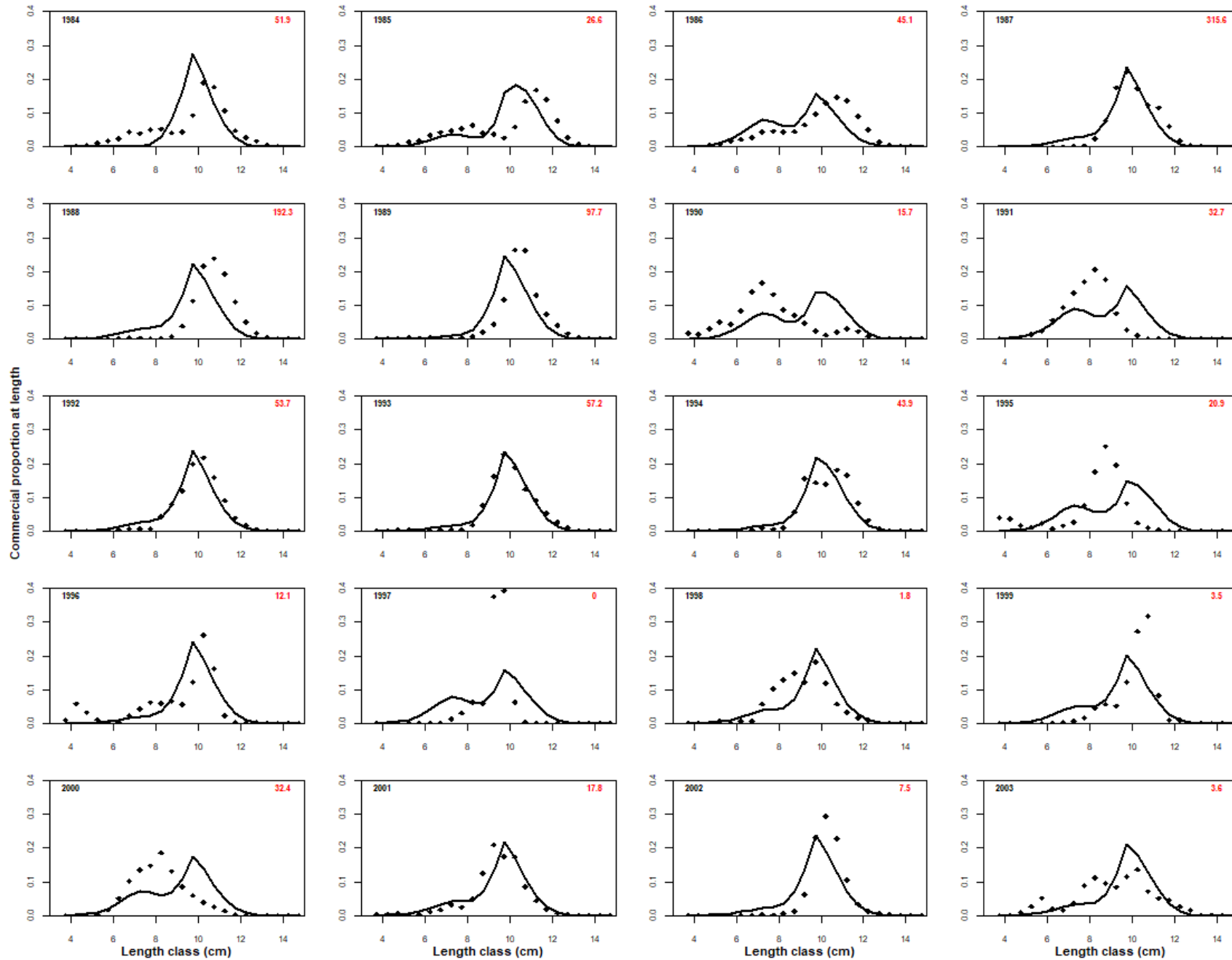




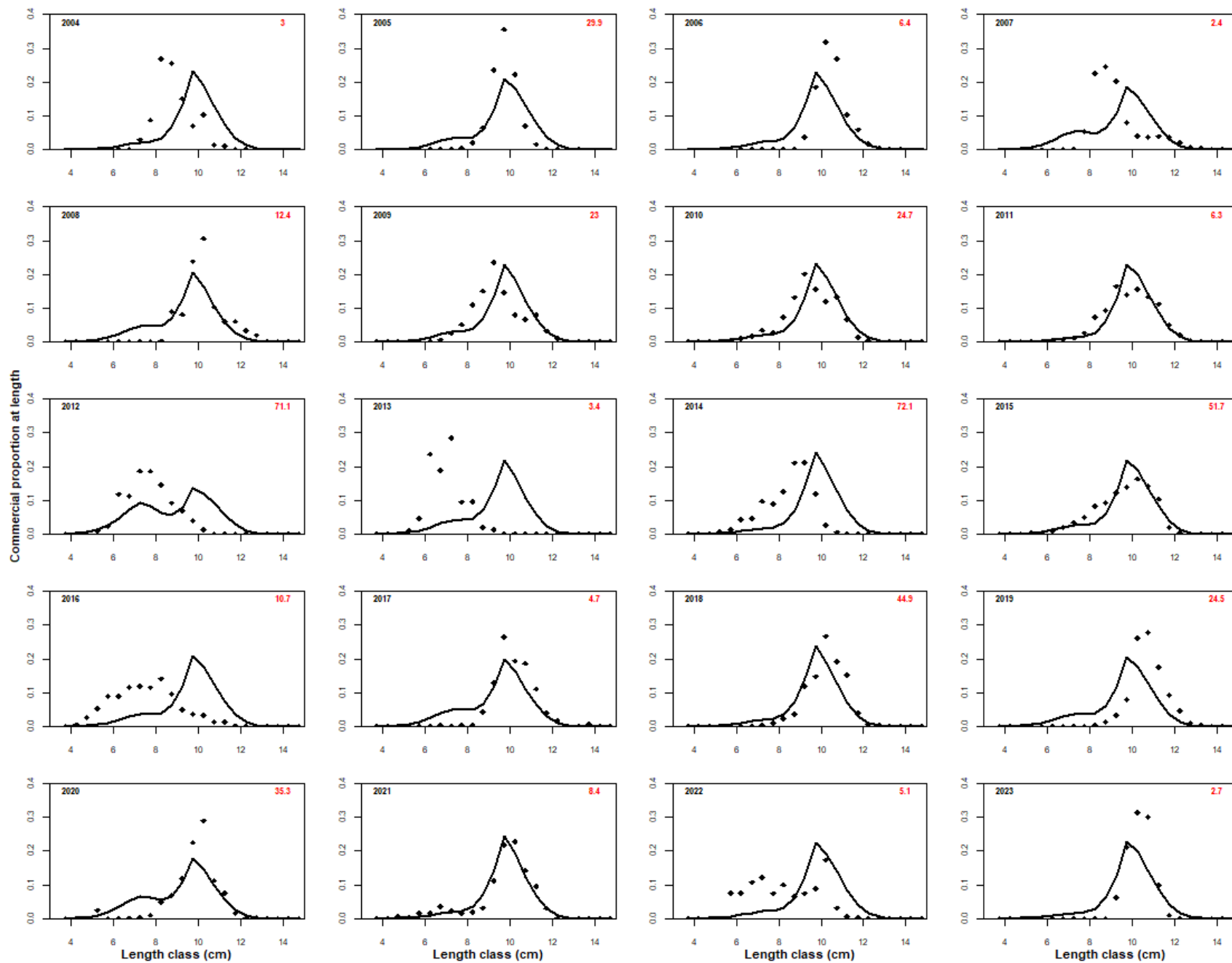


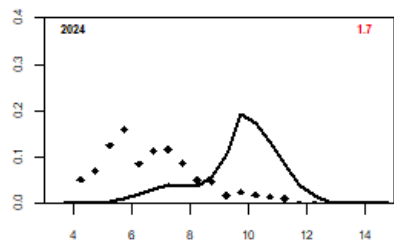
Appendix D: Model predicted and observed proportions-at-length in the quarterly commercial catches for A_{BH} .

Quarter 1

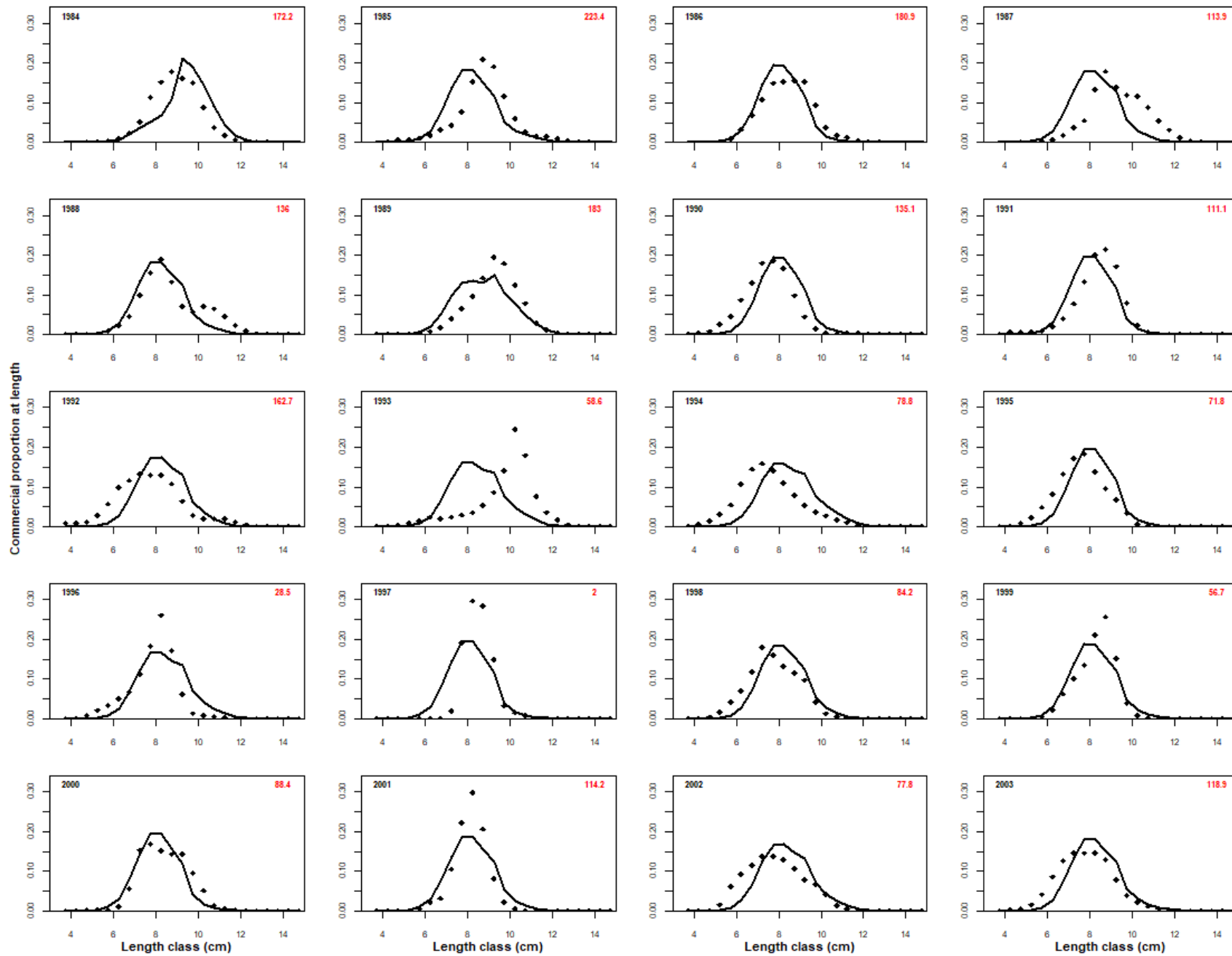


Quarter 1

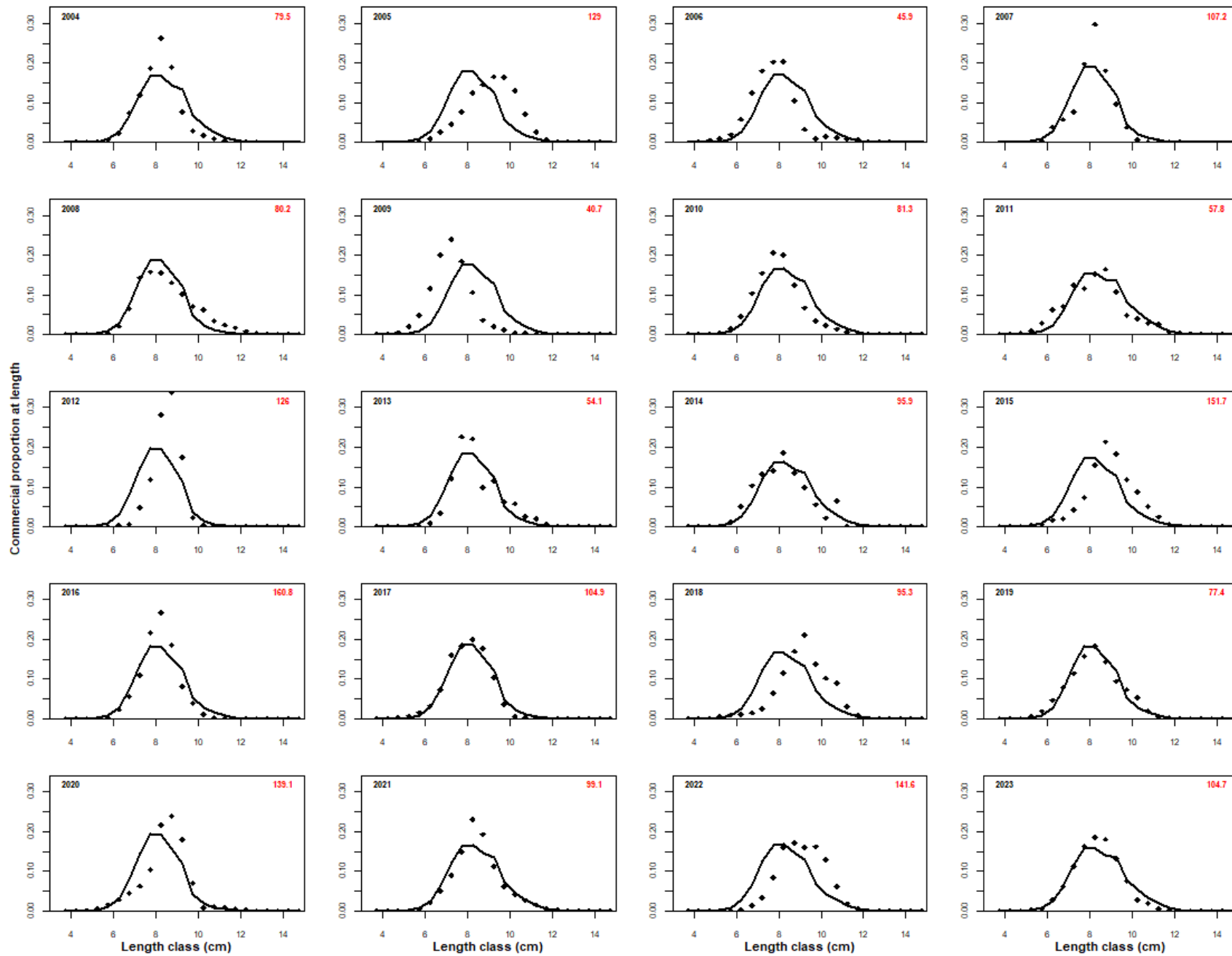


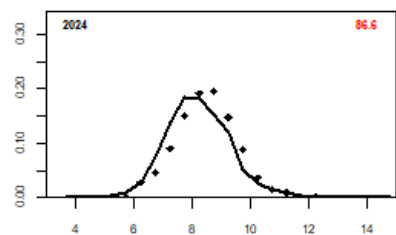


Quarter 2

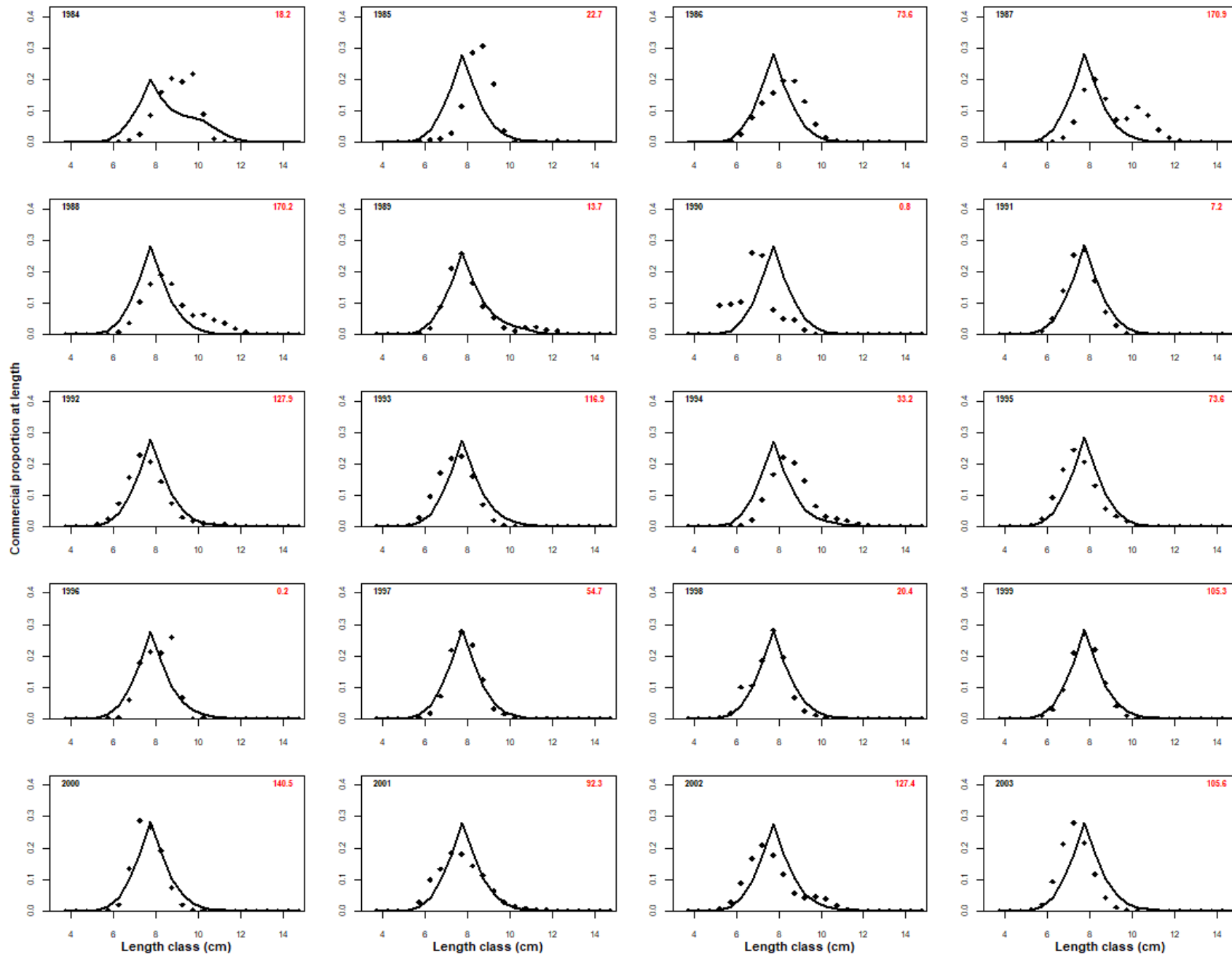


Quarter 2

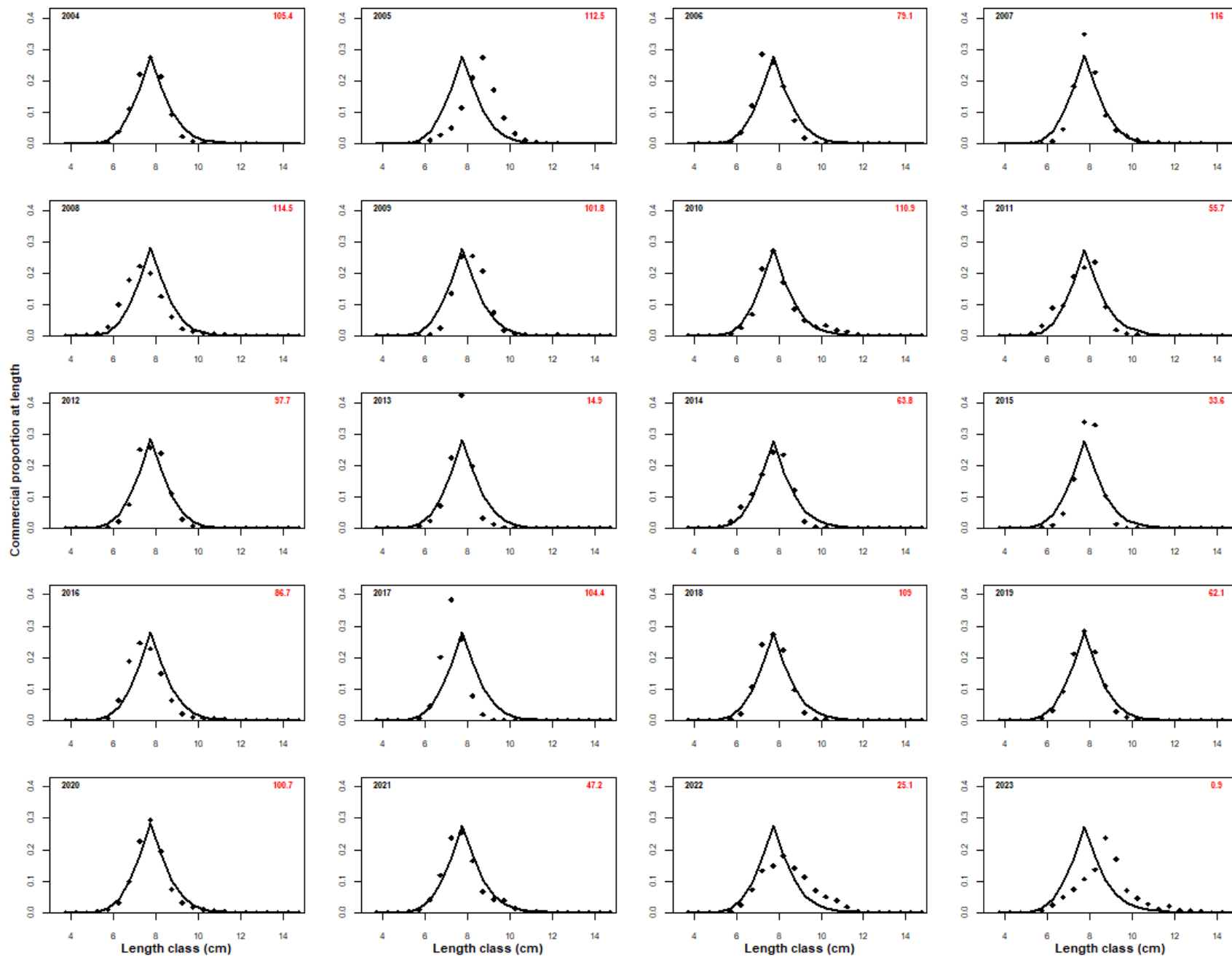


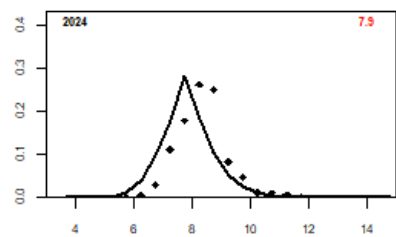


Quarter 3

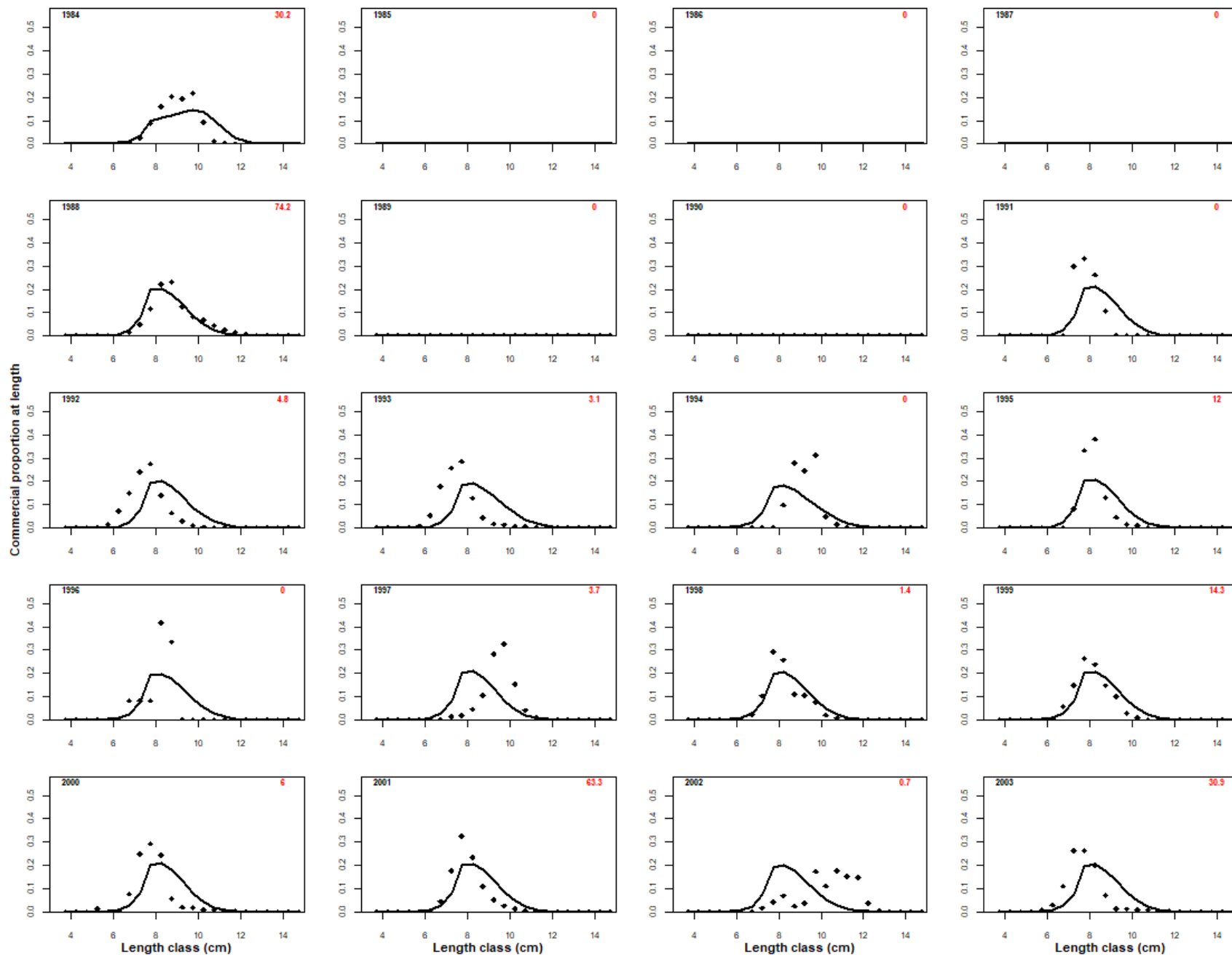


Quarter 3





Quarter 4



Quarter 4

